

The Trend-Cycle Connection*

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Abstract: Business cycles in emerging economies are more volatile than in advanced economies, and their long-run growth is lower and repeatedly interrupted by Sudden Stops. We argue that these facts are connected. Around a Sudden Stop, the trade balance reverses, driven by a compression of imports, especially capital goods. Because imported investment has a lasting effect on capital accumulation, its collapse generates persistent losses in output. Our model features an endogenous growth trend shaped by investment dynamics and financial conditions, allowing Sudden Stops to permanently affect the trend. We estimate the model using Argentine data from 1951 to 2015. The model reproduces the dynamics and frequency of Sudden Stops observed in the data. Productivity shocks account for most of the trend, but the imported-capital channel, amplified by financial frictions, is what makes Sudden Stops persistent. Without endogenous growth, the comovement that defines these crises would be about half as strong. In emerging markets, the trend and the cycle are deeply intertwined.

Keywords: emerging markets, real business cycle, trend shocks, financial frictions.

JEL Classification: F32, F34, F41.

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1 Introduction

Emerging economies' business cycles are far more volatile than those of advanced economies, while their long-run growth is lower and is interrupted by Sudden Stops far more often. These facts are usually studied in isolation. In this paper, we argue that they are connected: during Sudden Stops the growth rate collapses and output does not return to its pre-crisis path, so that the cycle shapes the trend. We develop a theory of the emerging-market business cycle in which the trend is partly endogenous, shaped by the accumulation of capital. In it, recurrent Sudden Stops tighten financial frictions and interrupt investment, generating the excess volatility and the long-run stagnation observed in Latin American economies.

We start by documenting the regularities that motivate our paper, first on a broad cross-section of emerging economies and then, at the country level, for Argentina, which is the object of our quantitative analysis. Around a Sudden Stop, the trade balance reverses sharply, and the reversal is driven by a compression of imports rather than by a rise in exports. While the role of imports is well known in the literature (see, for instance, [Alessandria et al., 2015](#); [Gopinath and Neiman, 2014](#)), we show that within imports, capital goods fall the most and are the slowest to recover, a pattern much milder in ordinary downturns. Moreover, across episodes, deeper collapses of imported capital are followed by slower recoveries of output. Capital goods are the component of imports that builds the productive capital stock. Their collapse is therefore a natural channel through which a Sudden Stop can have persistent effects on output growth, so that, in the sense of [Aguilar and Gopinath \(2007\)](#), the cycle becomes the trend. Because Sudden Stops are endogenous, however, aggregate data alone cannot establish this channel. We therefore build and estimate a structural model to measure it.

We build a dynamic stochastic general equilibrium model of a small open economy with three key features: (1) endogenous growth, (2) financial frictions in the corporate sector, and (3) reliance on imported investment. The trend has two components: a shock and an endogenous part that depends on the stock of capital, both domestic and imported. Financial frictions restrict firms' ability to invest during periods of distress, that is, a Sudden Stop, leading to sharp declines in domestic and imported investment that undermine the accumulation of capital and lower trend growth, while simultaneously improving the trade balance through import compres-

sion. This mechanism generates the core dynamics of [Aguiar and Gopinath \(2007\)](#) endogenously. Importantly, our model endogenizes the relationship between trend growth and the cycle by making capital accumulation, and in particular imported investment, responsive to financial conditions. This allows us to reconcile the observed macroeconomic volatility of emerging markets with the long-lasting effects of financial crises.

We estimate the model for Argentina over a long annual sample that runs from 1951 to 2015, a period that includes six Sudden Stops and large macroeconomic swings. Besides the standard observables, we use domestic investment, imported investment, and the price of imported investment, which discipline the imported-capital margin. The estimated financial frictions are much larger in the imported-capital sector than in the domestic one, making imported capital the margin most exposed to a tightening of external conditions. The model fits the key moments of the data, none of which are targeted, and reproduces the dynamics around Sudden Stops. It also generates Sudden Stops endogenously at a frequency of 7.9 percent, close to the 9.4 percent observed in the data, a moment the estimation does not target.

We use the estimated model to decompose the role of real and financial factors during normal times and Sudden Stops. It is uncontroversial that Sudden Stops affect the trend growth of the economy. Nevertheless, there is no agreement in the literature regarding the drivers of the emerging-market business cycle. On the one hand, [Aguiar and Gopinath \(2007\)](#) suggest that the main business cycle dynamics in emerging markets come from trend productivity shocks. On the other hand, [Garcia-Cicco et al. \(2010\)](#) suggest that the key feature behind the emerging-market business cycle is financial frictions. Our model provides a decomposition of the channels in a context of a partially endogenous trend. What favors productivity in the data is the tight comovement of output, consumption, and domestic investment, which productivity shocks generate. Financial shocks instead dominate imported investment and the external variables. These findings suggest that financial factors operate as relevant amplification channels of supply shocks in emerging economies. Endogenous growth also amplifies the defining comovement of Sudden Stops: without it, the negative correlation between output growth and the trade balance would be about half as strong.

Focusing on Sudden Stops, we quantify their long-run effect on the trend, episode by episode, and decompose its main drivers. Counterfactual economies in which we

remove capital from the trend, or freeze only imported capital, isolate the channel. Without the endogenous-capital term, financial and transitory shocks do not reach long-run growth at all: capital accumulation is the only way they affect the trend, and the endogenous component is what turns each crisis into a drawn-out decline of the trend. Shocks specific to the external variables, namely risk in the imported-capital sector and the price of imported investment, reach the trend mainly through imported capital, while domestic and aggregate shocks are transmitted mostly through domestic capital. Applied to Argentina’s history, the same decomposition attributes the level of the trend loss mainly to productivity, while imported capital plays a non-negligible role in transmitting the financial shocks. Our model suggests that Sudden Stops have a strong and persistent effect on the trend of the economy.

Our main contribution is to identify and quantify the channels through which Sudden Stops affect long-run growth. By tightening financial conditions and disrupting the accumulation of imported capital, a Sudden Stop lowers the capital stock and, through it, trend growth. Existing explanations for the persistent effects of financial crises emphasize domestic innovation, such as R&D and technology adoption (e.g., [Queralto, 2020](#)). We view the two mechanisms as complementary, each relevant for a different class of economies. In the emerging economies that suffer recurrent Sudden Stops, R&D spending is small, while imported capital goods account for a large share of imports and investment. Empirically, we document new facts on the composition of the import collapse during Sudden Stops, showing that imported capital is its deepest and slowest-recovering component and that the depth of its collapse predicts the speed of the output recovery. To do so, we build a long, consistent series of imported capital goods for a broad cross-section of emerging economies. Methodologically, we disaggregate capital and financial frictions and bring imported investment and its price into the estimation, which is what allows us to identify the imported-capital channel.

Related literature. Our paper contributes to the literature on business cycles in emerging economies. [Aguiar and Gopinath \(2007\)](#) show that trend shocks play a dominant role in these countries and that trend volatility accounts for their excess output volatility relative to developed economies. This view was challenged by [Garcia-Cicco et al. \(2010\)](#), who argue that financial frictions, not just trend shocks, are critical to understanding macroeconomic dynamics in emerging markets. These contributions initiated a growing literature, including [Chang and Fernández \(2013\)](#), [Seoane \(2016\)](#),

and [Miyamoto and Nguyen \(2017\)](#), among others, that examines the role of trend shocks and financial frictions in shaping emerging-market cycles. In most of these models, however, trend growth is exogenous and the financial friction is modeled as a reduced-form function of debt, output, or the terms of trade, so they focus on business cycle behavior and allow only limited feedback from the real economy to financial conditions.

Closely related to our financial block is the work of [Fernández and Gulan \(2015\)](#), who, like [Akinci \(2021\)](#), embed a Bernanke–Gertler–Gilchrist (BGG) financial accelerator in a small open emerging economy. They introduce a BGG financial contract into a small open economy business cycle model and estimate it on a panel of emerging economies that combines macroeconomic and financial data, showing that endogenous financial frictions reproduce the countercyclicality of country interest rates and the dynamics of corporate leverage that characterize these economies. We share this foundation and this empirical strategy. Our contribution is distinct in two respects. First, we disaggregate capital into domestic and imported components and allow the financial friction to differ across them, with monitoring costs that are substantially higher for imported capital. This asymmetry makes imported-capital accumulation the margin most exposed to a tightening of financial conditions. Second, and central to the paper, we embed this financial block in a model with endogenous growth, so that the disruption of imported capital during a Sudden Stop is transmitted to the economy’s trend. While [Fernández and Gulan \(2015\)](#) use financial frictions to account for the cyclical behavior of interest rates and leverage, we use them, together with the imported-capital margin and endogenous growth, to account for the persistent effects of Sudden Stops on long-run growth. Relative to both strands, growth is not only shaped by crises but also feeds back into the incidence of crises, a two-way link between cycle and trend.

Two earlier studies are particularly relevant to our approach. [Guerron-Quintana and Jinnai \(2019\)](#) examine whether the 2008 financial crisis had permanent effects on U.S. output, using a closed-economy model with financial constraints. [Queralto \(2020\)](#), which we mentioned before, analyzes the long-term effects of financial crises on productivity and innovation, focusing on the 1997 Korean episode. While both highlight the persistent consequences of financial disruptions, neither considers the role of imported capital goods, which, as we show in the next section, are distinctive for business cycle and Sudden Stop dynamics. Our account is complementary: we

develop an emerging-economy model with an endogenous trend and imported investment that locates the persistence of financial crises in the accumulation of imported capital rather than in domestic innovation, offering a more structural explanation of persistent post-crisis stagnation.

Our paper also relates to recent work on how financial frictions shape long-run growth. [Ottonello and Winberry \(2024\)](#) develop an endogenous growth model with heterogeneous firms in which financial frictions distort the allocation between investment and innovation and generate sizable losses in aggregate productivity, and [Bonciani et al. \(2023\)](#) show, in a model with endogenous growth, that financial crises cause slow recoveries that macroprudential policy can attenuate. This literature centers on domestic firms and closed or advanced economies. Instead, we emphasize a margin that is central for emerging economies, the accumulation of imported capital, and trace its transmission to the trend. Closest to our mechanism, [Cugat and Manera \(2024\)](#) construct an index of the technology embodied in a country’s imports and show that it predicts future growth in emerging markets, consistent with imported capital enabling domestic upgrading.

Finally, our paper relates to the literature on the interaction between endogenous growth and business cycles. [Comin \(2004\)](#) and [Comin and Gertler \(2006\)](#) investigate the medium-term macroeconomic cycle but abstract from structural features that are central to emerging markets, such as exposure to volatile capital flows and dependence on imported investment goods. A related strand explores how cyclical shocks can generate persistent effects: [Ates and Saffie \(2021\)](#), [Matsumoto \(2022\)](#), and [Benguria et al. \(2022\)](#) analyze growth volatility in emerging markets, while [Anzoategui et al. \(2019\)](#) and [Bianchi et al. \(2019\)](#) examine related dynamics in advanced economies. Our contribution to this branch is a quantitative decomposition of the drivers of growth volatility in an emerging-market setting, with an emphasis on the long-run impact of financial crises through investment dynamics.

2 Some facts during Sudden Stops

This section documents the empirical regularities of Sudden Stops that motivate our theory and discipline its estimation. We establish four facts. First, emerging economies grow more slowly and more erratically than advanced economies. Second, they experience Sudden Stops far more often. Third, around a Sudden Stop the trade

balance reverses sharply, and this reversal is driven by a compression of imports rather than by a rise in exports. Fourth, capital goods are the main driver of that import compression: they collapse by more than the other import categories and recover more slowly, and by far more than in ordinary downturns. This last fact matters because capital goods are the component that builds the productive capital stock, so their collapse during a Sudden Stop makes the episode persistent, weighing on growth for several years and, ultimately, on the trend. We establish these facts on a broad cross-section of emerging economies and then, using Argentina as a laboratory with recurrent episodes, confirm them at the country level.

2.1 Emerging economies

Emerging economies combine low average growth with high volatility. Table 1 reports, for the period 1960–2018, the population-weighted mean growth rate of real per-capita output, its volatility, and the number of Sudden Stops, for three groups: emerging small open economies, advanced small open economies, and Latin American countries, a subset of emerging countries.¹ Emerging economies grew on average 1.9 percent per year, against 2.2 percent in advanced small open economies, and their output growth was about 80 percent more volatile. Latin American countries show a more pronounced version of the same pattern: they display the lowest average growth, 1.7 percent, and the highest incidence of Sudden Stops.

¹Following [Uribe and Schmitt-Grohé \(2017\)](#), we classify a country as emerging if its average PPP-converted GDP per capita over 1990–2017 is between 3,000 and 25,000 US dollars of 2005, and as advanced if it is above that range. Among emerging economies we keep small open economies from Latin America and the Caribbean, Europe, Central Asia, and the Middle East and North Africa; among advanced economies we use the sample of developed small open economies of [Garcia-Cicco et al. \(2010\)](#). The full country lists are in the note to the table.

Table 1: Growth and Sudden Stop statistics (1960–2018)

	Mean growth rate (%)	Growth volatility (%)	Number of Sudden Stops
Emerging SOEs	1.93	4.22	91 (3)
<i>of which: Latin America</i>	<i>1.73</i>	<i>3.71</i>	<i>39 (4)</i>
Developed SOEs	2.15	2.33	11 (1)

Notes: Population-weighted cross-sectional averages of country-level statistics. The number of Sudden Stops is the total across countries in each group, with the average per country in parentheses. Latin American countries are a subset of the emerging small open economies (SOEs). *Emerging SOEs*: Albania, Algeria, Argentina, Antigua and Barbuda, Barbados, Belize, Bolivia, Brazil, Bulgaria, Chile, Colombia, Costa Rica, Cuba, Dominica, Dominican Republic, Ecuador, Egypt, El Salvador, Grenada, Guatemala, Guyana, Honduras, Iran, Jordan, St. Lucia, Morocco, Mexico, Panama, Paraguay, Peru, Tunisia, Turkey, Uruguay, and Venezuela. *Latin America*: Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, Mexico, Paraguay, Peru, Uruguay, and Venezuela. *Developed SOEs*: Australia, Austria, Belgium, Canada, Denmark, Finland, Iceland, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, and Switzerland.

We define a Sudden Stop as a year in which real GDP falls by at least 2 percent and the trade-balance-to-output ratio rises by at least 2 percentage points, following Seoane and Yurdagul (2019). As Table 1 shows, emerging economies, and Latin American ones in particular, suffer this type of crisis far more often than advanced economies. Together with their lower average growth and higher volatility, this suggests that Sudden Stops leave a persistent imprint on economic growth. These patterns motivate a model in which the trend is not purely exogenous but is partially affected by the occurrence of Sudden Stops.

To characterize the dynamics of these episodes, we construct an event study around each Sudden Stop.² Figure 1 reports the cross-episode averages.³

²We set the year of the Sudden Stop as period $t = 0$ and follow each observable over the five years before and after the episode. An episode is discarded if the five-year window falls outside the sample or if another Sudden Stop occurred fewer than five years earlier. This procedure yields 59 emerging-market Sudden Stops.

³Output growth and the trade variables are annual log changes of the real per-capita series. Imported capital is imports of capital goods (Broad Economic Categories 41 and 521), from UN Comtrade and CEPII’s BACI database, spliced within country across the 1996 change in trade classification (Appendix A.1); interest payments are external-debt interest (World Bank International Debt Statistics) over GDP, shown as the change from the pre-crisis year. Because of data availability, panels use every episode for which the variable is observed: output growth and net exports on 59 episodes (1965–2010), export and import growth on 50, imported-capital growth on 43 (1976 onward), and interest payments on 37.

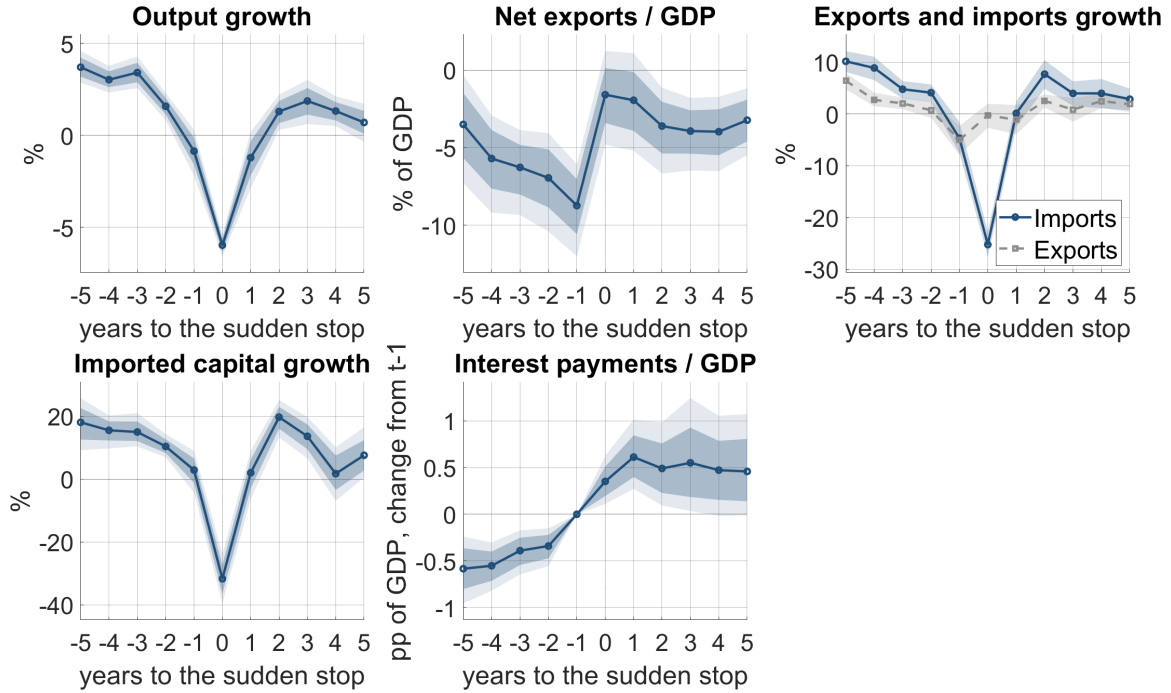


Figure 1: Sudden Stop dynamics in emerging countries

Note: Cross-episode means around emerging-market Sudden Stops (period 0). Solid lines: means; shaded areas: 68% (dark) and 90% (light) country-block bootstrap bands (68% only in the exports-imports panel). Output growth and export/import growth are annual log changes of the real per-capita series ($\times 100$); net exports and interest payments are ratios to GDP, the latter shown as the change from the pre-crisis year.

Output contracts sharply and does not fully catch up. Output growth falls by about 8 percentage points relative to its pre-crisis average at impact and recovers only gradually, so the level of output remains below the path it would have followed absent the crisis, producing a permanent scar in the economy.

The trade balance follows a V-pattern: it deteriorates in the years leading to the crisis and reverses sharply in the Sudden Stop year. This reversal is driven to a large extent by a compression of imports rather than by a rise in exports.⁴ In volume terms, imports collapse by about 25 percent at the crisis while exports are essentially flat, consistent with the view that export capacity cannot be expanded quickly.

⁴Appendix A.3 decomposes the change in net exports over GDP into the contributions of the export- and import-to-GDP ratios. The fall in the imports-to-GDP ratio accounts for about three-fifths of the reversal at impact and the rise in the exports-to-GDP ratio for the remaining two-fifths. This ratio decomposition understates the role of imports, because the contraction of GDP raises the export-to-output ratio even when export volumes are flat.

The interest-payments-to-output ratio rises at the crisis, by about half a percentage point of GDP, and remains elevated for several years. This is consistent with the tightening of external financing conditions that accompanies a Sudden Stop, and it is a variable our estimation later exploits to identify financial shocks.

The imported capital panel of Figure 1 previews the margin our theory emphasizes: the growth rate of imported capital collapses by roughly 30 percent at the Sudden Stop. Figure 2 places this collapse in context by splitting imports into capital, intermediate materials, and consumption goods, and by comparing Sudden Stops with ordinary downturns.

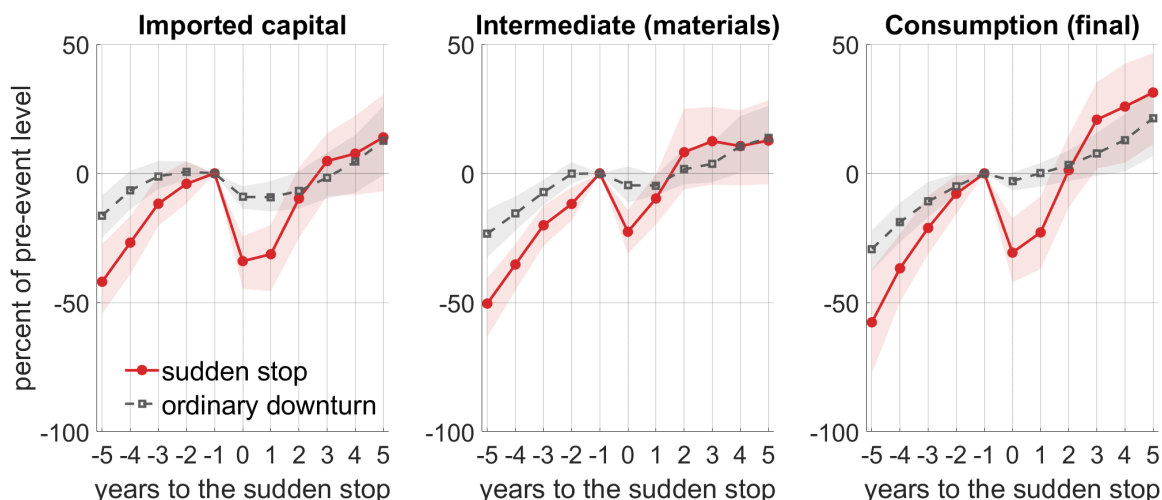


Figure 2: Import categories around Sudden Stops (red) and ordinary downturns (gray)

Note: Cumulative log level of each import category relative to the pre-event year, around emerging-market Sudden Stops (37 episodes, red) and around ordinary downturns, defined as all years of negative real GDP growth outside Sudden Stops in the same sample (128 downturn-years, 23 countries, gray). Shaded areas are 95 percent country-block bootstrap bands.

Figure 2 conveys two points about imported capital. First, its collapse is specific to Sudden Stops. Capital-goods imports fall by about 34 percent at impact in a Sudden Stop, against only about 9 percent in an ordinary downturn. Consumption and intermediate imports also fall by more in Sudden Stops than in ordinary downturns, but the contrast is starkest for capital goods, so the deep collapse of imported capital is a feature of these crises rather than of normal recessions. Second, within the import

bill, capital goods are the component that matters for long-run growth. They record the deepest fall on impact, about 34 percent, comparable to consumption goods, about 31 percent, and larger than intermediate materials, about 23 percent, and, together with the other categories, account for the reversal in imports.⁵ What sets capital goods apart is not the size of their fall alone but their economic role: they are the imports that add to the productive capital stock, so their collapse is the one that can be transmitted to the trend. Consumption imports, even though they fall sharply, do not build capital and therefore cannot affect long-run growth. This is the margin our model isolates.

Our mechanism places imported capital, rather than domestic innovation, at the center of long-run growth in these economies. Figure 3 shows why. In countries that suffer recurrent Sudden Stops, research and development (R&D) spending is very low: the median country spends about 0.3 percent of GDP on R&D. Imported capital, by contrast, is a large share of the import bill, between 13 and 29 percent across our sample. Where innovation intensity is this limited, the accumulation of capital, much of it imported, is a more plausible engine of the trend than domestic R&D. Our mechanism is in this sense complementary to that of [Queralto \(2020\)](#), who emphasizes R&D, and better suited to the economies we study.

⁵Appendix A.4 decomposes the year-over-year change in total imports into an imported-capital component and all other imports. Imported capital falls more than proportionally, accounting for roughly 30 percent of the total-import collapse at the Sudden Stop, more than its share of total imports.

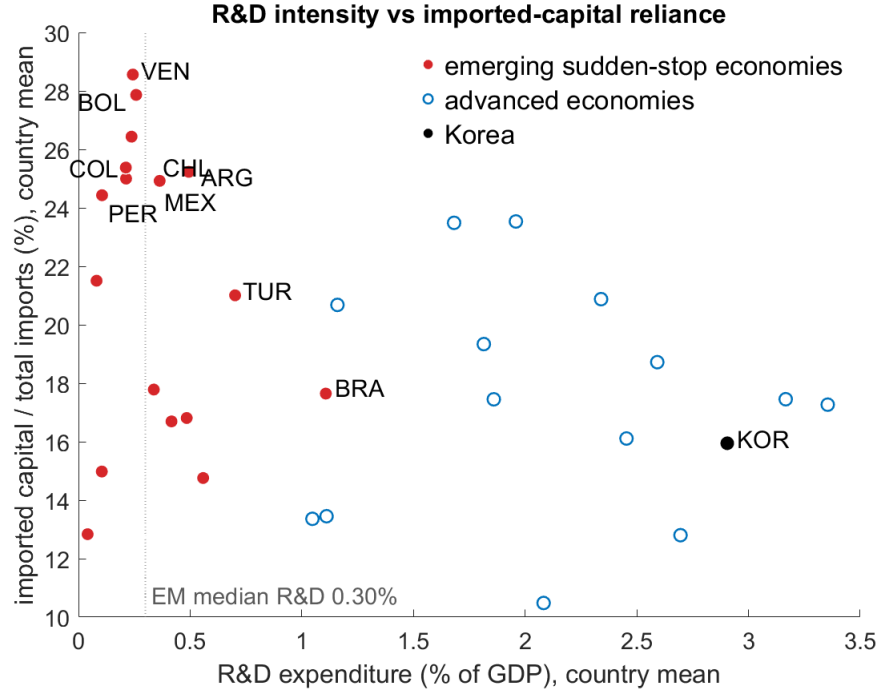


Figure 3: R&D intensity versus imported-capital reliance, by country

Note: Country means over the sample period. Horizontal axis: R&D expenditure over GDP (World Bank). Vertical axis: imported capital goods (BEC 41 and 521) over total imports. Solid circles: emerging economies in the Sudden Stop sample; open circles: advanced economies; black: Korea. The dotted line marks the sample median R&D share.

We rely on the structural model to analyze a causal relationship between Sudden Stops, growth, and the role of imported capital. However, Figure 4 presents suggestive evidence that imported capital is a key variable for growth in emerging economies in the presence of Sudden Stops. The vertical axis measures the persistence of the Sudden Stop, defined as the number of years real GDP per capita takes to regain its pre-crisis level. In the left panel this level is the pre-crisis peak, that is, the highest level of output per capita in the years leading up to the Sudden Stop. In the right panel, it is the average level of output per capita over the five years before the onset.⁶ The horizontal axis measures the collapse of imported capital during the Sudden Stop, defined as the percent fall in imports of capital goods from their pre-crisis peak, the

⁶A recovery time of zero indicates that output per capita never fell below the chosen pre-crisis level within the recovery window, so no positive recovery time is recorded. These cases arise under the average benchmark (right panel), which is a less demanding threshold than the pre-crisis peak, and are absent under the peak benchmark (left panel).

highest level in the five years before the Sudden Stop, to their trough.⁷ There is a clear positive relationship between the two, with a correlation of about 0.5 under both benchmarks: economies whose imported capital collapses more deeply take longer to recover.

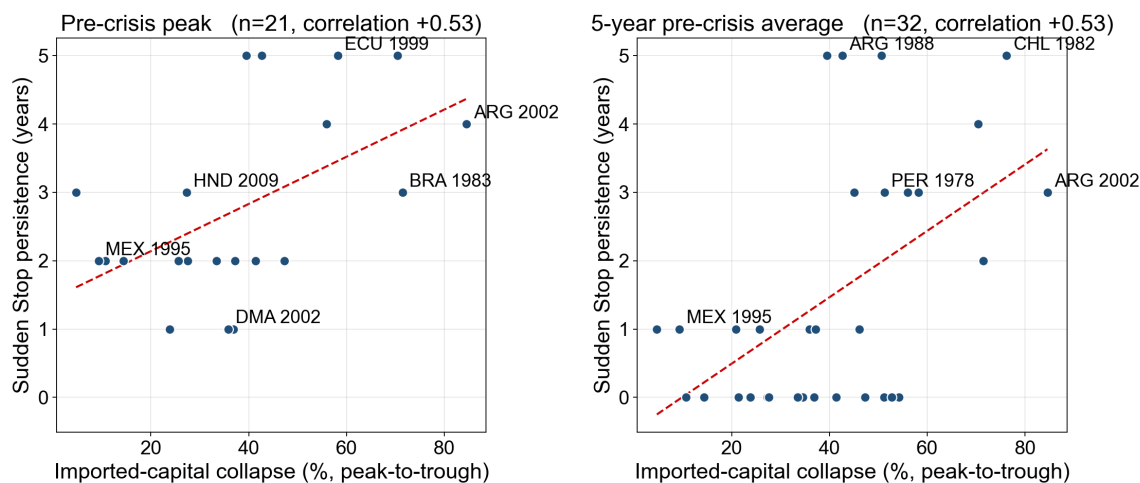


Figure 4: Imported-capital collapse and the persistence of Sudden Stops

Note: Each point is an emerging-market Sudden Stop. Horizontal axis: the percent fall in imports of capital goods from their pre-crisis peak, the highest level in the five years before the Sudden Stop, to their trough. Vertical axis: the number of years real GDP per capita takes to regain its pre-crisis level within a five-year window; the benchmark is the pre-crisis peak in the left panel and the average of the five preceding years in the right panel. The dashed line is a linear fit.

The figure is based on the 39 Sudden Stop episodes for which the peak-to-trough collapse of imported capital can be measured, that is, those with imported-capital data both before and during the crisis. The vertical axis requires the economy to regain its pre-crisis output level within a five-year window, which we impose so that recovery windows do not overlap with subsequent Sudden Stops, so not every episode enters each panel: 21 recover within five years under the pre-crisis-peak benchmark (left panel) and 32 under the pre-crisis-average benchmark (right panel). The omitted episodes are the most severe: those that do not regain the pre-crisis peak within five years have a mean imported-capital collapse of 47 percent, against 38 percent for

⁷Measuring this collapse from the year before the Sudden Stop, rather than from the pre-crisis peak, yields correlations of 0.47 and 0.42 for the two benchmarks, so the positive relationship is not sensitive to this choice.

those that do. Because these deeper and slower episodes would lie in the upper-right region of the chart, their omission makes the positive relationship a conservative lower bound.

2.2 Argentina’s stylized facts

Argentina is a natural laboratory for these facts. It exhibits the regularities documented above and, having experienced many Sudden Stops over our sample, it offers repeated episodes to study at higher resolution.⁸ Figure 5 reports the dynamics around the six Argentine episodes.

The Argentine episodes reproduce the cross-country pattern. Imported investment collapses by about 54 percent at the crisis and is the main driver of the fall in imports. As in the cross-country sample, it is the imported-investment margin, the one that builds the capital stock, that dominates the import collapse. The interest-payments-to-output ratio rises sharply in the crisis year and contributes to the trade-balance reversal. This variable is useful for two reasons. First, and unlike Calvo et al. (2006), our definition of a Sudden Stop does not use interest rates, which are unavailable for Argentina before the 1980s. The sharp rise of interest payments at the crisis is therefore independent evidence that we are identifying the right episodes. Second, it carries information about financial conditions that our estimation later uses to identify financial frictions and shocks.

⁸We use the national accounts of the *Instituto Interdisciplinario de Economía Política de Buenos Aires*, Instituto Interdisciplinario de Economía Política de Buenos Aires (IIEP, UBA-CONICET) (2018), which separately measure investment in domestic and in imported transport and equipment goods; total imported investment is the sum of the two. Applying the methodology described above to Argentine data for 1951–2015, we obtain six Sudden Stops: 1959, 1976, 1982, 1988, 1995, and 2002.

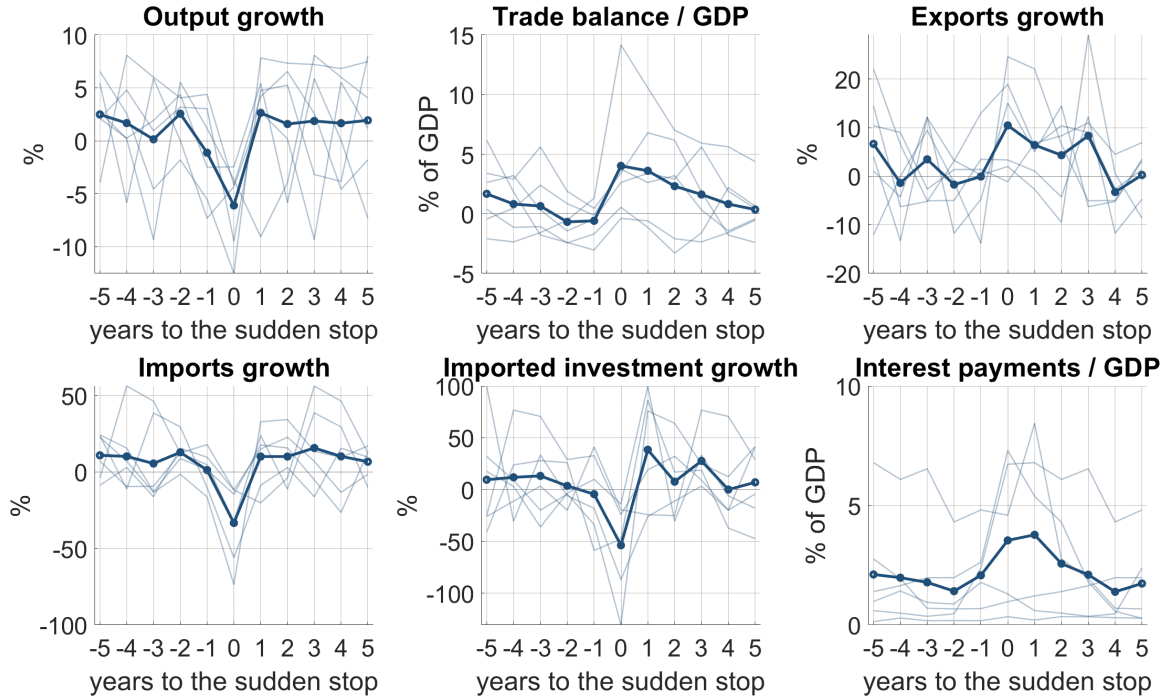


Figure 5: Sudden Stop dynamics in Argentina

Note: Period 0 is the year of the Sudden Stop. Thin lines: the six individual episodes (1959, 1976, 1982, 1988, 1995, 2002); thick line: the cross-episode mean. Output, exports, imports, and imported investment growth are annual growth rates of the real per-capita series; the trade-balance and interest-payment ratios are in percent of GDP.

2.3 Taking stock

Both the cross-country evidence and the Argentine case highlight the same features: emerging economies grow less and more erratically, suffer Sudden Stops far more often, and in each episode the trade balance reverses through a compression of imports led by capital goods, the component that builds the capital stock and plausibly transmits the crisis to long-run growth.

These facts highlight the importance of imported capital. Because capital goods build the productive capital stock, their deep and slow-to-recover collapse in each Sudden Stop plausibly leaves a lasting imprint on growth. Because Sudden Stops are endogenous, aggregate time-series and cross-sectional evidence cannot by itself establish this causal link. We therefore build a structural model designed to generate these facts, in which the trend is endogenous and Sudden Stops feed into it through the ac-

cumulation of domestic and imported capital. The model endogenizes the hypothesis of [Aguiar and Gopinath \(2007\)](#): the cycle is the trend in emerging economies, but the trend is itself shaped by Sudden Stops. We estimate the model on Argentina, a canonical emerging country with repeated Sudden Stops. Although we do not target it, the estimated model generates Sudden Stops endogenously at a frequency similar to the data, which is our second fact. The recurrence of these episodes, in turn, generates the excess volatility of growth and the persistent output losses that each crisis leaves behind, at the center of the trend-cycle connection. Taking the model to the data is thus our test of the theory: by estimating it we ask whether a structure disciplined by the third and fourth facts assigns to imported capital the central role that our reading of the evidence suggests, and we use it to measure the contribution of real and financial shocks to the interaction between cycle and trend. In what follows, we present the theory and the strategy to take it to the data.

3 The model

The model is a small open economy with endogenous growth and financial frictions, populated by households, final goods producers, capital goods producers, domestic investment goods producers, entrepreneurs, and the government.

The central mechanism works as follows. During Sudden Stops, the trade balance adjusts through a sharp compression of imports, in particular imports of investment goods. Because imported capital goods contribute to the economy's growth trend, this compression lowers long-run growth. Financial frictions amplify this channel: the tightening of credit to entrepreneurs during a Sudden Stop reduces investment persistently. Consequently, financial crises leave permanent scars in the economy, so that to a large extent the cycle is the trend.

The mechanism hinges on the specification of the trend, which has both an exogenous component and an endogenous component driven by the aggregate capital stock, including imported capital. By compressing imported investment and depressing capital accumulation, Sudden Stops therefore have a permanent effect on long-run growth. When the endogenous component is shut down, the model nests the exogenous trend of [Aguiar and Gopinath \(2007\)](#) and financial shocks have no permanent effect. This connects to the debate between [Aguiar and Gopinath \(2007\)](#), who em-

phasize trend shocks, and [Garcia-Cicco et al. \(2010\)](#), who argue for financial frictions. Our model accommodates both forces and lets them interact.

Beyond the core mechanism, two modeling choices respond to empirical concerns. First, capital goods production is split into an imported and a domestic sector. Their relative prices behave very differently in the data, the former exhibiting a pronounced secular trend and the latter being stationary, and the disaggregated investment series allow us to discipline each sector separately. Second, each sector has its own entrepreneur, subject to sector-specific financial frictions and to a stochastic dispersion of idiosyncratic productivity shocks. This dispersion provides a source of financial disturbance independent of aggregate productivity movements. Given the quantitative nature of our analysis, we view these features as essential rather than secondary. We describe each agent in detail below.

3.1 Households

The economy is populated by a continuum of measure one of identical households, characterized by a representative agent. Households own all firms in the economy, supply labor to both the final goods and the domestic investment goods sector, and borrow from international financial markets. We use GHH preferences ([Greenwood et al., 1988](#)) augmented with external habit formation. GHH preferences eliminate the wealth effect on labor supply, a property that helps the model match the relationship between interest rates and output in emerging markets. The habit parameter α governs consumption inertia. Households do not internalize the habit formation effect on aggregate consumption, $\tilde{C}$, when making individual decisions.

Their optimization problem consists of maximizing their lifetime utility,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \nu_t \beta^t \left[\left(C_t - \alpha \tilde{C}_{t-1} - X_{t-1} \frac{h_{f,t}^{\omega_f}}{\omega_f} - X_{t-1} \frac{h_{d,t}^{\omega_d}}{\omega_d} \right)^{1-\sigma} \frac{1}{1-\sigma} \right] \quad (1)$$

subject to an infinite set of budget constraints, $\forall t$,

$$C_t + D_t R_t = W_{d,t} h_{d,t} + W_{f,t} h_{f,t} + D_{t+1} + \Lambda_t. \quad (2)$$

Households choose consumption C_t , external borrowing D_{t+1} and labor supply $h_{d,t}$, $h_{f,t}$. We assume a specific labor supply for the domestic investment sector, $h_{d,t}$,

and the final goods sector, $h_{f,t}$, with their corresponding wages $W_{d,t}$ and $W_{f,t}$. Parameters ω_d and ω_f characterize the corresponding labor supply elasticities. Treating the two labor types as imperfect substitutes introduces, in reduced form, a cost of reallocating labor across the consumption and investment sectors. Without it, labor would move frictionlessly between sectors and the model would smooth the sharp sectoral reallocation that a Sudden Stop produces. R_t is the domestic interest rate. X_{t-1} is the trend of the economy, which we explain in detail in the following sections. Λ_t denotes transfers and profits received by households every period t :

$$\Lambda_t = \Pi_{ki,t} + \Pi_{kd,t} + \Pi_{Id,t} + T_{i,t} + T_{d,t} - S_t.$$

Here $\Pi_{ki,t}$, $\Pi_{kd,t}$, and $\Pi_{Id,t}$ denote profits of imported capital and domestic capital and investment goods producers. $T_{i,t}$ and $T_{d,t}$ are net real transfers of imported and domestic capital, paid to incoming entrepreneurs and received from exiting ones. S_t denotes lump-sum taxes paid to the government. We provide a detailed description of profits and transfers in the following section.

Finally, ν_t is a preference shock that follows an AR(1) process:

$$\ln \nu_{t+1} = \rho_\nu \ln \nu_t + \epsilon_{t+1}^\nu; \quad \epsilon_t^\nu \sim N(0, \sigma_\nu^2); \quad |\rho_\nu| < 1. \quad (3)$$

The country interest rate combines an exogenous risk-free component, an exogenous spread shock, and an endogenous spread that depends on aggregate debt and output. The endogenous component captures the empirically documented counter-cyclicality of interest rate spreads in emerging markets (Neumeyer and Perri, 2005; Uribe and Yue, 2006). The household takes the interest rate as given, since it does not internalize the effect of its individual borrowing decisions on the country spread. The exogenous components include risk-free interest rate shocks, $R_{f,t}$, and spread shocks, μ_t . The latter represents exogenous variations in the interest rate that the domestic economy has to pay for its debt and is independent of its fundamentals. The timing for the debt and spread shocks follows the timing in Justiniano and Preston (2010), where spread shocks affect contemporaneously the cost of repaying the debt. The two shocks are separately identified in the estimation: $R_{f,t}$ is tied to the observed world real interest rate, while μ_t is identified from the interest-payments-to-output ratio.

The interest rate is $R_t = R_{o,t-1}e^{\mu_t-1}$, with

$$R_{o,t} = R^* + R_{f,t} + \psi_D \left[\exp \left(\frac{\tilde{D}_{t+1} + \tilde{B}_{t+1}}{X_t} - (\bar{d} + \bar{b}) \right) - 1 \right] + \psi_Y \left[\exp \left(\frac{Y_t}{X_{t-1}} - \bar{y} \right) - 1 \right]. \quad (4)$$

We assume that the endogenous spread has two parts: the first depends on deviations of detrended debt (from households and firms) to the average debt level. Here $\tilde{D}_{t+1}$ is aggregate household debt and $\tilde{B}_{t+1}$ is aggregate entrepreneurial (corporate) debt, $\tilde{B}_{t+1} = \tilde{B}_{i,t+1} + \tilde{B}_{d,t+1}$, which we introduce when we describe entrepreneurs below. The parameter that measures this debt elasticity of the interest rate is ψ_D , which is assumed to be positive since a higher debt level is associated with higher default risk. The second part depends on deviations of detrended output to the average output in the economy. The parameter ψ_Y allows us to capture the fact that the interest rate may fall when output is growing. The representative household does not internalize the effect of her decisions on the country's interest rate, which is affected by aggregate variables ($\tilde{D}$, $\tilde{B}$, and GDP Y_t).

R^* is the average interest rate, and $\bar{d}, \bar{b}, \bar{y}$ are steady-state values. We assume $R_{f,t}$ and μ_t follow a zero mean AR(1) process in logs:

$$\ln R_{f,t+1} = \rho_{R_f} \ln R_{f,t} + \epsilon_{t+1}^{Rf}; \quad \epsilon_t^{Rf} \sim N(0, \sigma_{R_f}^2); \quad |\rho_{R_f}| < 1, \quad (5)$$

$$\ln \mu_{t+1} = \rho_\mu \ln \mu_t + \epsilon_{t+1}^\mu; \quad \epsilon_t^\mu \sim N(0, \sigma_\mu^2); \quad |\rho_\mu| < 1. \quad (6)$$

3.2 Capital goods producers

Capital goods are produced in two symmetric, perfectly competitive sectors, one for imported capital (sector I) and one for domestic capital (sector D). We keep the sectors distinct for two reasons. First, the relative price of imported investment goods has a pronounced downward trend in the data, while the price of domestic investment is stationary, and a single-sector model could not reproduce this asymmetry. Second, the two-sector structure lets us match disaggregated data on domestic and imported capital, which in turn helps identify the endogenous growth mechanism. The specification follows [Christiano et al. \(2010\)](#) and [Fernández-Villaverde \(2010\)](#).

Consider a representative producer in sector $j = \{I, D\}$. It combines the predetermined installed-capital stock $K_{j,t}$ with new investment $I_{j,t}$ to produce next period's capital stock $K_{j,t+1}$, which it sells to entrepreneurs in sector j at the price $q_{j,t}$. There is a single installed-capital good per sector: new and used capital are the same good and trade at the same price $q_{j,t}$, with no separate market or price for used capital. The producer's choice variables are $I_{j,t}$ and $K_{j,t+1}$, while $K_{j,t}$ is predetermined. All producers are competitive and take $q_{j,t}$ and $P_{j,t}$ as given.

The two sectors differ only in the relative price of investment goods, $P_{j,t}$. In the imported sector, this price inherits a deterministic trend, $P_{i,t} = p_{i,t}\Xi_{t-1}$, where Ξ_{t-1} is the deterministic trend that captures the secular decline of the imported investment price relative to domestic goods observed in the data, and $p_{i,t}$ is stationary and follows an AR(1) process with mean $\bar{p}$,

$$\ln(p_{i,t+1}/\bar{p}) = \rho_p \ln(p_{i,t}/\bar{p}) + \epsilon_{t+1}^p; \quad \epsilon_t^p \sim N(0, \sigma_p^2); \quad |\rho_p| < 1. \quad (7)$$

In the domestic sector, the relative price does not grow, so $P_{d,t} = p_{d,t}$.

For $j = \{I, D\}$, the optimization problem of a representative producer is

$$\max_{K_{j,t+1}, I_{j,t}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{\tilde{\lambda}_t}{\lambda_0} \left(q_{j,t} K_{j,t+1} - q_{j,t} K_{j,t} (1 - \delta_{kj}) - P_{j,t} I_{j,t} \right),$$

subject to

$$K_{j,t+1} = K_{j,t} (1 - \delta_{kj}) + I_{j,t} - \Phi_j \left(\frac{K_{j,t+1}}{K_{j,t}} \right) K_{j,t}. \quad (8)$$

As is standard in the literature, we assume quadratic capital adjustment costs, which take the following functional forms for domestic capital,

$$\Phi_d \left(\frac{K_{d,t+1}}{K_{d,t}} \right) = \frac{\phi_d}{2} \left(\frac{K_{d,t+1}}{K_{d,t}} - \bar{g} \right)^2,$$

and for imported investment capital,

$$\Phi_i \left(\frac{K_{i,t+1}}{K_{i,t}} \right) = \frac{\phi_i}{2} \left(\frac{K_{i,t+1}}{K_{i,t}} - \frac{\bar{g}}{\bar{g}_{\Xi}} \right)^2.$$

Here, $\bar{g}$ is the average growth rate of the economy, and $\bar{g}_{\Xi}$ is the average growth rate of the imported investment price. The two costs differ because the price of domestic investment is stationary while $P_{i,t}$ is allowed to grow, as in the data. The different

growth rates in investment prices generate different growth rates in investment between sectors and then in domestic and imported capital, as we show in Section B.3 of the Appendix.

Substituting the law of motion (8) into the objective, profits can be written as $(q_{j,t} - P_{j,t})I_{j,t} - q_{j,t}\Phi_j(K_{j,t+1}/K_{j,t})K_{j,t}$, which shows that $K_{j,t}$ enters only as the pre-determined scale of the adjustment cost, not as an additional decision. Because adjustment costs make the installed-capital price $q_{j,t}$ differ from the investment price $P_{j,t}$, the undepreciated stock carried into $t + 1$ is the same good revalued at $q_{j,t+1}$. This revaluation appears in the producer's optimality condition (Appendix B.3) and in the entrepreneur's gross return defined below.

3.3 Entrepreneurs

There are two types of entrepreneurs, one for each capital sector. This two-type structure is not merely a symmetry assumption. It lets the model assign different financial friction parameters to the imported and domestic capital sectors, in particular different monitoring costs μ_{ki} and μ_{kd} and different processes for the dispersion of the idiosyncratic shock. This captures the empirically plausible idea that financing conditions for imported and domestic capital need not be the same. The contract between entrepreneurs and foreign lenders follows the costly-state-verification framework of Christiano et al. (2014). Because the idiosyncratic productivity shock is the entrepreneur's private information, the lender must pay a monitoring cost to verify the true state in default. This gives rise to an endogenous external finance premium that rises when net worth falls and amplifies the investment response to financial shocks. We focus here on the economic structure of the contracting problem, leaving the formal derivation of the optimal contract to Appendix B.

Every period t , an entrepreneur N_j buys capital $K_{j,t+1}^N$ from a capital producer in sector j , refurbishes it using a stochastic linear technology, and rents it to the final goods producer. In order to purchase the capital, entrepreneurs can use their net worth, $N_{j,t+1}$, or issue defaultable debt, $B_{j,t+1}^N$, lent by an international financial intermediary. Hence,

$$q_{j,t}K_{j,t+1}^N = N_{j,t+1} + B_{j,t+1}^N. \quad (9)$$

The effective capital the entrepreneur obtains is $\omega_{j,t+1}^N K_{j,t+1}^N$, where ω_j^N is an idiosyncratic shock for each entrepreneur N , type j . This shock is independently drawn across time, type, and entrepreneurs.

We assume ω_j^N is log-normal with unit mean, $E_t(\omega_{j,t+1}^N) = 1$, and time-varying dispersion $\sigma_{\omega,t}^j$, for $j = \{i, d\}$. The unit-mean normalization ties the location parameter to the dispersion,

$$\mathbb{E}_t \omega_{j,t+1}^N = e^{\mu_{\omega_j,t+1} + \frac{1}{2}\sigma_{\omega,t+1}^{2,j}} = 1 \implies \mu_{\omega_j,t+1} = -\frac{1}{2}\sigma_{\omega,t+1}^{2,j},$$

so a rise in dispersion lowers the location of the distribution and shifts mass into its left tail, raising the fraction of entrepreneurs that default. Following [Christiano et al. \(2014\)](#) and [Fernández-Villaverde \(2010\)](#), the dispersion varies stochastically over time and follows

$$\log(\sigma_{\omega,t}^j) = (1 - \rho_\sigma^j) \log(\mu_\sigma^j) + \rho_\sigma^j \log(\sigma_{\omega,t-1}^j) + \eta_\sigma^j \epsilon_{\sigma^j,t}, \quad \epsilon_{\sigma^j,t} \sim N(0, 1).$$

The shock to the dispersion of ω_j^N can be interpreted as a financial-risk shock that operates independently of aggregate productivity. A higher dispersion implies riskier entrepreneurs, which raises the external finance premium and compresses investment even when aggregate TFP is unchanged.

The transformation of raw capital into effective units takes one period. Then entrepreneurs buy $K_{j,t+1}^N$ at period t , but rent the capital services at period $t+1$, at rate $r_{j,t+1}$. At the end of period $t+1$, entrepreneurs are left with $(1 - \delta_{kj})\omega_{j,t+1}^N K_{j,t+1}^N$ and sell it to a capital producer in sector j .

The entrepreneur's return per unit of capital purchased in t is $R_{j,t+1}\omega_{j,t+1}^N$, and the average return per unit invested in period t and sector j is

$$R_{j,t+1} = \frac{r_{j,t+1} + q_{j,t+1}(1 - \delta_{kj})}{q_{j,t}}. \quad (10)$$

Under this contract, the foreign lender is risk neutral and, in default, pays a monitoring cost to verify the entrepreneur's realized productivity and then seizes the remaining assets, recovering a fraction $1 - \mu_{kj}$ of the return in sector j . Let $\bar{\omega}_{j,t+1}$ denote the productivity threshold below which the entrepreneur defaults and F the

log-normal CDF, and define

$$\begin{aligned}\Gamma(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) &= \bar{\omega}_{j,t+1} (1 - F(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j)) + G(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j), \\ G(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) &= \int_0^{\bar{\omega}_{j,t+1}} \omega dF(\omega, \sigma_{\omega,t}^j),\end{aligned}$$

which are, respectively, the lender's expected gross share of entrepreneurial earnings and the expected earnings from projects that default, which accrue to the lender gross of the monitoring cost μ_{kj} . The entrepreneur chooses the leverage ratio $\varsigma_{j,t} = B_{j,t+1}/N_{j,t+1}$ and the default threshold to maximize expected net worth subject to the lender's zero-profit condition. The resulting optimality condition is

$$\begin{aligned}\mathbb{E}_t \left[\frac{R_{j,t+1} q_{j,t} K_{j,t+1}}{R_{t+1} N_{j,t+1}} (1 - \Gamma(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j)) \right] = \\ \mathbb{E}_t \left[\frac{1 - F(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j)}{1 - F(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) - \mu_{kj} \bar{\omega}_{j,t+1} F(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j)} \right].\end{aligned}\quad (11)$$

Because the idiosyncratic shock is i.i.d. across entrepreneurs within a sector, all entrepreneurs of type j make the same decisions. We therefore drop the index N and work with the sector aggregates B_j and K_j .

Given such a contract, the law of motion of entrepreneurial net worth is given by

$$N_{j,t+1} = \frac{1}{1 + e^{-\bar{\gamma}^e}} \left[R_{j,t} q_{j,t-1} K_{j,t} - R_t B_{j,t} - \mu_{kj} \int_0^{\bar{\omega}_{j,t}} \omega dF(\omega) R_{j,t} q_{j,t-1} K_{j,t} \right] + w_j^e X_{t-1}, \quad (12)$$

where $\bar{\gamma}^e$ regulates the survival rate of entrepreneurs. Exiting entrepreneurs transfer their net worth to households, and these fund incoming entrepreneurs by transferring w_j^e . The net of these operations is reflected in the term $T_{j,t}$ observed in the households' budget constraint, which is given by

$$T_{j,t} = \left(1 - \frac{1}{1 + e^{-\bar{\gamma}^e}} \right) V_{j,t} - w_j^e X_{t-1}. \quad (13)$$

Here, $V_{j,t}$ is the net worth before the fraction of entrepreneurs regulated by $\bar{\gamma}^e$ leaves the market and is given by

$$V_{j,t} = R_{j,t} q_{j,t-1} K_{j,t} - R_t B_{j,t} - \mu_{kj} \int_0^{\bar{\omega}_{j,t}} \omega dF(\omega) R_{j,t} q_{j,t-1} K_{j,t} \quad (14)$$

Notice that we are imposing the same survival rate for entrepreneurs in the domestic and imported capital sectors.

3.4 Domestic investment producer

Domestic investment goods $I_{d,t}$ are produced using labor $h_{d,t}$.⁹ We assume a decreasing returns to scale technology:

$$I_{d,t} = \tilde{a}_t^d (h_{d,t})^\rho,$$

with $0 < \rho < 1$ and $\tilde{a}_t^d = \bar{a}_d a_{d,t} X_t$. $\bar{a}_d$ is a constant that determines the average productivity level, and $a_{d,t}$ is a specific TFP shock that follows an AR(1) in logs, correlated with the final goods TFP shock through a common innovation. We specify its process together with that of a_t below. Notice that we assume that the TFP in the domestic investment sector follows the same growth trend as in the final goods production.

The optimization problem of this firm is the following:

$$\max_{h_{d,t}} \Pi_{I_{d,t}} = p_{d,t} \tilde{a}_t^d (h_{d,t})^\rho - W_{d,t} h_{d,t}$$

The firm pays wages equal to its marginal product, and profits are distributed to the households.

3.5 Final goods producer

The final goods sector is where the paper's central mechanism is embedded. Domestic and imported capital enter the production function as imperfect substitutes through a CES Armington aggregator. Because of this imperfect substitutability, a contraction in imported capital, for instance when a Sudden Stop tightens the credit available to entrepreneurs in the imported sector, cannot be fully offset by domestic capital, so aggregate output falls. The key additional feature is that the economy's growth trend X_t depends endogenously on the aggregate capital stock. The Sudden Stop therefore also lowers the long-run growth of the economy. The parameter η governs

⁹Modeling this sector with labor only is a deliberate simplification adopted for tractability: it keeps the two capital-production sectors comparable and focuses the model on the imported-capital margin, where our mechanism operates.

the relative weight of the exogenous and endogenous components of the trend. When $\eta = 1$ the trend is fully exogenous and the model nests [Aguiar and Gopinath \(2007\)](#). When $\eta < 1$ the trend has an endogenous component that allows financial frictions to propagate into long-run growth.

The competitive firm in this sector maximizes the following profit function:

$$\Pi_{f,t} = Y(K_{i,t}, h_{f,t}, K_{d,t}, a_t, X_t) - r_{d,t}K_{d,t} - W_{f,t}h_{f,t} - r_{i,t}K_{i,t}$$

This firm solves an intratemporal problem and pays each input its marginal product.

The production function is the following:

$$Y_t = a_t(X_t h_{f,t})^\gamma K_t^{1-\gamma}, \quad (15)$$

where K_t represents the total capital services. The Armington aggregator for capital is given by

$$K_t = (a_1 K_{d,t}^{\mu_1} + (1 - a_1) (\Xi_{t-1} K_{i,t})^{\mu_1})^{\frac{1}{\mu_1}}.$$

The CES specification implies imperfect substitutability between domestic and imported capital, and it is similar to the one used in [Mendoza and Yue \(2012\)](#) and [Park \(2017\)](#).

As discussed before, Ξ_{t-1} is the deterministic trend in the imported investment price $P_{i,t}$. The previous expression implies that the aggregate capital K_t grows at the trend of the economy X_{t-1} . X_t is given by

$$X_t = \Gamma_t^\eta \left[\left(a_1 \tilde{K}_{d,t}^{\mu_1} + (1 - a_1) \left(\Xi_{t-1} \tilde{K}_{i,t} \right)^{\mu_1} \right)^{\frac{1}{\mu_1}} \right]^{1-\eta} \quad (16)$$

and the growth rate of the trend is $g_{x,t} = \frac{X_t}{X_{t-1}}$. Γ_t is an exogenous stochastic trend such that $\frac{\Gamma_t}{\Gamma_{t-1}} = g_t$, which follows an AR(1),

$$\ln(g_{t+1}/\bar{g}) = \rho_g \ln(g_t/\bar{g}) + \epsilon_{t+1}^g; \quad \epsilon_t^g \sim N(0, \sigma_g^2); \quad |\rho_g| < 1, \quad (17)$$

while $\tilde{K}_{d,t}$ and $\tilde{K}_{i,t}$ are aggregate (non-internalized) capital inputs that drive the endogenous component of the trend.

The expression in Eq. (16) nests the trend specification of [Aguiar and Gopinath \(2007\)](#) as a special case when $\eta = 1$, in which growth is fully exogenous. We gen-

eralize this by introducing an endogenous component, inspired by AK-type growth models, whereby the trend depends on the level of productive capital, both imported and domestic. Intuitively, the long-run growth rate is partly determined by capital accumulation, which allows financial frictions and Sudden Stops to influence the trend through their effect on investment. This mechanism endogenizes the “cycle is the trend” hypothesis and provides a reconciliation with the evidence in [Garcia-Cicco et al. \(2010\)](#), who emphasize the role of financial frictions in emerging markets.¹⁰

The productivity a_t is a mean-reverting productivity shock and follows an AR(1) process in logs:

$$\ln a_{t+1} = \rho_a \ln a_t + \rho_{a,ad} \epsilon_{t+1}^{a,ad} + \epsilon_{t+1}^a; \quad \epsilon_t^a \sim N(0, \sigma_a^2); \quad |\rho_a| < 1, \quad (18)$$

where $\epsilon_{t+1}^{a,ad}$ is a common shock, $\epsilon_t^{a,ad} \sim N(0, 1)$, that affects the TFP in both the production of domestic investment and the production of the consumption goods sector. That is, this shock allows us to account for a potential correlation between the mean-reverting components of TFP. Symmetrically, the sectoral TFP shock of the domestic investment sector, $a_{d,t}$ (introduced above), follows

$$\ln a_{d,t+1} = \rho_{ad} \ln a_{d,t} + \rho_{ad,a} \epsilon_{t+1}^{a,ad} + \epsilon_{t+1}^{ad}; \quad \epsilon_t^{ad} \sim N(0, \sigma_{ad}^2); \quad |\rho_{ad}| < 1, \quad (19)$$

so that the common innovation $\epsilon^{a,ad}$ enters the two mean-reverting TFP components with loadings $\rho_{a,ad}$ and $\rho_{ad,a}$.

3.6 Government

We follow [Garcia-Cicco et al. \(2010\)](#) and model government consumption as a domestic spending shock s_t that reverts to a constant share $\bar{s}$ of output and follows an AR(1) in its innovations:

$$s_{t+1} = (1 - \rho_s) \bar{s} y_{t+1} + \rho_s s_t + \epsilon_{t+1}^s; \quad \epsilon_t^s \sim N(0, \sigma_s^2); \quad |\rho_s| < 1,$$

¹⁰We adopt this specification for tractability and identification purposes. Because we have disaggregated data on domestic and imported capital stocks, we can discipline the endogenous growth mechanism directly. Alternative growth models, such as those driven by R&D or learning-by-doing, are conceptually appealing but not feasible here due to the lack of consistent historical data on innovation inputs for Argentina.

where $s_t = \frac{S_t}{X_{t-1}}$. Households finance this government spending through lump-sum taxes. We include this shock to align with the existing literature and to make our definition of output in line with that of the data. After estimation, we find that this shock plays a minor role in model dynamics.

3.7 Balance of payments

Consolidating the budget constraints of all agents yields the economy's aggregate resource constraint. The model features two distinct types of foreign debt, household debt D_t , used to smooth consumption, and corporate debt $B_t = B_{i,t} + B_{d,t}$, used by entrepreneurs to finance capital purchases in both sectors.

Combining the household budget constraint (2), the financing identity (9), the entrepreneurial transfers (13)–(14), the intermediate producers' profits, and the optimality conditions of the final goods firm yields the aggregate resource constraint (the consolidation is detailed in Appendix B):

$$GDP_t = C_t + p_{d,t}I_{d,t} + P_{i,t}I_{i,t} + S_t + TB_t, \quad (20)$$

where TB_t is the trade balance:

$$TB_t = R_t D_t - D_{t+1} + R_t(B_{i,t} + B_{d,t}) - (B_{i,t+1} + B_{d,t+1}).$$

GDP_t , the variable we observe in the data, is

$$GDP_t = Y_t + p_{d,t}I_{d,t} - \mu_{ki} \int_0^{\bar{\omega}_{i,t}} \omega dF(\omega) R_{i,t} q_{i,t-1} K_{i,t} - \mu_{kd} \int_0^{\bar{\omega}_{d,t}} \omega dF(\omega) R_{d,t} q_{d,t-1} K_{d,t}. \quad (21)$$

Moreover, we define the net interest rate payments to the rest of the world as follows:

$$rb_{y,t} = (R_t - 1) \frac{D_t + B_t}{GDP_t}$$

We present the complete set of equilibrium equations in the appendix.

4 Empirical strategy

We log-linearize the stationarized equilibrium conditions of the model and estimate it with a Bayesian strategy using annual data for Argentina for 1951 to 2015 from [Instituto Interdisciplinario de Economía Política de Buenos Aires \(IIEP, UBA-CONICET\) \(2018\)](#). We calibrate some of the parameters to match first-order moments of the data and to align with standard values in the existing literature. We estimate the remainder of the parameters with Metropolis-Hastings and loose priors.

Table 2 presents the values of constrained parameters with their corresponding source or target. The CRRA coefficient, σ , which defines the curvature of the period utility function, is set to 2, and the depreciation rates (δ_{ki} , δ_{kd}) to 10 percent. The discount factor β equals 0.94 to target an annualized interest rate of 8 percent in the steady state. We set $\bar{g}$ to 1.01, the average gross growth rate of output per capita, and $\bar{g}_\Xi$ to 0.9756, the growth rate of the imported investment price. The average trade-balance-to-output ratio is 1.4 percent, as in Argentina in the period under study. γ and ρ , the coefficients of labor in the production function of the final good and domestic investment, are set equal to 2/3. We set the Armington weight of domestic capital, a_1 , to 0.62, in line with [Mendoza and Yue \(2012\)](#)'s calibration for imported inputs. The preference parameter ω_d is set equal to 1.2, as in [Akinci \(2021\)](#).¹¹

We set the ratio B/N in each sector to the average leverage ratio in industrial investment for firms in Argentina in the period 2007–2016.¹² We fix $\bar{\gamma}^e$ to match the average percentage of surviving firms per year, equal to 89 percent in Argentina from 2007 to 2016.

Finally, we set the relative price of imported investment in steady state to the average value in the data, $\bar{p} = 1.22$.

¹¹We also tried to estimate this parameter, and in most of the trials, the chain of this parameter approached 1. To avoid convergence issues, we decided to fix it to a low value that aligns with the literature.

¹²To the best of our knowledge, information about this ratio is available only since 2007.

Table 2: Calibrated Parameters

Parameter	Value	Definition	Source or Target
σ	2	Risk aversion	Standard
γ	2/3	Labor coefficient in final good	Standard
ρ	2/3	Labor coefficient in domestic investment	Standard
β	0.94	Discount factor	8% average interest rate
δ_{kj}	0.10	Depreciation rate ($j = i, d$)	Standard
$\bar{\gamma}^e$	2.068	Entrepreneur survival rate	11% annual firm exit, Argentina 2007–2016
w_d^e	0.034	Transfer to entrepreneurs (domestic capital)	$(B/N)_d = 0.6$, Argentina 2007–2016
w_i^e	≈ 0	Transfer to entrepreneurs (imported capital)	$(B/N)_i = 0.6$, Argentina 2007–2016
$\bar{p}$	1.22	Avg. relative price of imported investment	Argentina 1951–2015
$\bar{g}_\Xi$	0.9756	Growth rate of imported investment price	Argentina 1951–2015
$\bar{g}$	1.01	Growth rate of output per capita	Argentina 1951–2015
$\bar{s}$	0.1144	Government spending to output ratio	Argentina 2007–2016
a_1	0.62	Armington weight on domestic capital	Mendoza and Yue (2012)
ω_d	1.2	Preference parameter	Akinci (2021)

We include eight observables: (1) GDP growth (g_y); (2) private consumption growth (g_c); (3) domestic investment growth (g_{i_d}); (4) imported investment growth (g_{i_i}); (5) the trade-balance-to-output ratio (tb_y); (6) the growth rate of the relative price of imported capital investment goods to the GDP deflator (g_{p_i}); (7) the world (U.S.) real risk-free interest rate (R_f); and (8) the ratio of net interest rate service on the foreign net asset position to output (rb_y). In all cases, we work in per capita

terms and take natural logs, except for tb_y and rb_y . We incorporate the latter variable with R_f to identify financial frictions and shocks in the estimation.¹³

In the estimation, we add measurement errors to all observables. We plot the described time series in the appendix. There, we can see that the variable g_{pi} has a negative mean equal to -0.024 during the period, showing the negative trend in imported capital investment prices, as in the model.

The estimated parameters and the prior distributions are in Table 3. We consider loose priors because, given the complexity of the likelihood function, the estimation with flat priors tends to work poorly, as many estimates would hit the parameter bounds.

5 Estimation results

Table 3 presents the prior information, the posterior mean and median, and high probability density intervals (HPDI) of 10 percent and 90 percent for each estimated parameter. Posterior distributions are in similar orders of magnitude as in the existing literature. The HPDI of the debt elasticity of the interest rate, ψ_D , ranges from 0.02 to 0.14, smaller than the posterior mean in Garcia-Cicco et al. (2010), yet large enough to be quantitatively relevant for the dynamics. The model includes two features that explain the lower interest rate debt elasticity: first, the role of ψ_Y (absent in Garcia-Cicco et al. (2010)) and, second, the existence of additional financial frictions. The elasticity of the interest rate to output, ψ_Y , has a negative posterior mean equal to -1.31, supporting the hypothesis that financial frictions tend to relax during the expansive part of the cycle. Our estimates suggest that imported investment is subject to higher screening costs: the posterior mean of μ_{ki} is 0.27, about twice the standard calibration for this parameter in the case of the US. However, there seems to be a lower degree of financial friction in the domestic investment sector, where μ_{kd} takes a posterior mean of 0.03.¹⁴

A key parameter in our analysis is η , as it characterizes the importance of the endogenous component in the growth rate of this economy: the lower this parameter

¹³Due to data availability, our price deflators are Fisher chained indexes. Our original data for the NIPA accounts are nominal, so we can be consistent with the model and deflate output and consumption by using the GDP deflator. We deflate domestic investment and imported investment by using their own deflators. We provide a full description of the data treatment in the appendix.

¹⁴See Bernanke et al. (1999) and many others who calibrate this parameter to 0.12.

is, the more relevant is the endogenous component of the trend and the less important is the shock. The estimation places substantial mass around medium to large values of this parameter. Its posterior mean equals 0.69, and it has a high probability density interval of between 0.46 and 0.96.

Table 3: Priors and estimation results

	Dist.	Prior				Posterior			
		LB	UB	Mean	s.d.	Mean	Median	10%	90%
μ_σ^i	IG		2	1.00	0.25	0.82	0.81	0.66	0.97
μ_σ^d	IG		2	1.00	0.25	0.89	0.86	0.59	1.18
σ_g	IG			0.02	0.025	0.03	0.03	0.02	0.04
σ_a	IG			0.02	0.025	0.01	0.01	0.01	0.02
σ_{ad}	IG			0.02	0.025	0.01	0.01	0.01	0.02
σ_μ	IG			0.02	0.025	0.07	0.07	0.06	0.08
η_σ^i	IG			0.60	1.00	1.64	1.49	0.71	2.54
η_σ^d	IG			0.60	1.00	0.42	0.36	0.16	0.69
σ_{R_f}	IG			0.02	0.025	0.02	0.02	0.01	0.02
σ_ν	IG			0.02	0.025	0.08	0.08	0.05	0.13
σ_p	IG			0.02	0.25	0.21	0.21	0.18	0.24
σ_s	IG			0.02	0.025	0.01	0.01	0.01	0.01
ρ_g	Beta			0.30	0.10	0.41	0.41	0.24	0.58
ρ_a	Beta			0.30	0.10	0.47	0.47	0.34	0.60
ρ_{ad}	Beta			0.30	0.10	0.50	0.50	0.36	0.64
ρ_μ	Beta			0.30	0.10	0.30	0.30	0.20	0.40
ρ_σ^i	Beta			0.30	0.10	0.54	0.55	0.38	0.70
ρ_σ^d	Beta			0.30	0.10	0.30	0.30	0.14	0.46
ρ_{R_f}	Beta			0.30	0.10	0.48	0.48	0.36	0.59
ρ_ν	Beta			0.30	0.10	0.34	0.33	0.15	0.52
ρ_p	Beta			0.50	0.10	0.71	0.71	0.62	0.79
ρ_s	Beta			0.20	0.10	0.29	0.28	0.12	0.46
$\rho_{a,ad}$	Normal			0.20	0.50	0.02	0.02	0.01	0.03
$\rho_{ad,a}$	Normal			0.20	0.50	0.04	0.04	0.03	0.06
ϕ_i	Gamma			3.00	2.00	6.78	6.67	5.10	8.36
ϕ_d	Gamma			3.00	2.00	5.48	5.43	3.16	7.69
ψ_D	Normal	0.0	10.0	1.50	2.00	0.08	0.08	0.02	0.14
ψ_Y	Normal	-5.0	0.0	-1.00	0.25	-1.31	-1.31	-1.59	-1.04
μ_{ki}	Normal	0.0	1.0	0.12	0.10	0.27	0.27	0.13	0.41
μ_{kd}	Normal	0.0	1.0	0.12	0.10	0.03	0.03	0.00	0.07
ω_f	Normal	1.0	7.0	2.00	1.00	2.96	2.88	2.03	3.87
μ_1	Normal		0.91	0.50	2.00	0.80	0.80	0.72	0.91
α	Beta			0.25	0.10	0.13	0.13	0.06	0.19
η	Beta			0.50	0.25	0.69	0.70	0.46	0.96

Note: Posterior distributions from a random walk Metropolis-Hastings algorithm of 990,000 draws per chain across two chains, discarding the first half of each chain as burn-in.

The Armington curvature parameter μ_1 takes a posterior mean of 0.80, corresponding to an imperfect but quite high elasticity of substitution between domestic and imported capital, equal to $1/(1 - \mu_1) = 5$. The fact that imported and domestic inputs are substitutes aligns with the results of [Mendoza and Yue \(2012\)](#) and [Park \(2017\)](#), among others.

Table 4: Second-order moments

	g_y	g_c	g_{i_d}	g_{i_i}	tb_y	rb_y
Standard deviations (in %)						
Model	5.3	6.1	12.5	45.9	4.3	2.7
Data	5.2	6.8	12.9	41.3	3.2	2.1
Correlation with g_y						
Model	1.00	0.91	0.94	0.23	-0.18	-0.10
Data	1.00	0.91	0.92	0.62	-0.20	-0.35
Correlation with tb_y						
Model	-0.18	-0.26	-0.16	-0.07	1.00	0.71
Data	-0.20	-0.28	-0.22	-0.22	1.00	0.59

Note: Theoretical moments obtained by evaluating the parameters at their posterior means, imposing measurement errors' standard deviation equal to zero.

Table 4 shows the main second-order moments from the data and the model. The model fits the data: it produces the volatility ranking and a strong negative correlation between output growth and the trade balance, capturing the main stylized facts observed in emerging economies, as described in [Aguiar and Gopinath \(2007\)](#). This exercise is a test for the model, given that none of the numbers in the table are targeted during the estimation. It also generates the right volatility of the novel

variables, the growth rate of imported investment, and the interest-rate-payments-to-GDP ratio.¹⁵

Given the results in this section, we consider this model an adequate laboratory to study the anatomy of the business cycle dynamics and Sudden Stops in emerging countries.

6 Quantitative results

This section studies the main quantitative features of the model. We start by revisiting the business cycle drivers in emerging markets. Then we focus on how financial frictions operate as transmission channels of the shocks.

6.1 Drivers of the business cycle

Table 5 reports the variance decomposition. Trend shocks are a secondary driver of consumption, output, and investment, but they are the main driver of the trade-balance-to-output and debt-service-to-output ratios.¹⁶ This is in line with [Garcia-Cicco et al. \(2010\)](#) and [Akinci \(2021\)](#), where transitory shocks explain most of the variance of output, consumption, and domestic investment growth.¹⁷ Financial shocks (spread, risk-free rate, and risk) and the imported-investment price mainly explain imported investment, and to a lesser extent the trade balance, the interest-payments ratio, and consumption, while leaving output unaffected. The preference shock plays no role for output growth, but it accounts for about 10 percent of consumption growth

¹⁵In contrast to the standard approach in the literature, for instance, that of [Aguiar and Gopinath \(2007\)](#), [Garcia-Cicco et al. \(2010\)](#), and even [Chang and Fernández \(2013\)](#) or [Akinci \(2021\)](#), our framework brings a new set of identifying information to the estimation, as we include domestic and imported investment and imported investment price data as observables. The dynamics of these variables over the business cycle and during Sudden Stops are pronounced and allow us to better characterize the trade balance adjustment. Producing the right moments is, hence, challenging.

¹⁶Being a key driver of the debt-service-to-output ratio is an important feature of trend shocks because it suggests that this shock can play a central role in over-borrowing and external default phenomena, both of which are outside the scope of this paper but are considered in [Seoane and Yurdagul \(2019\)](#) and [Aguiar and Gopinath \(2006\)](#), respectively.

¹⁷In this table, the contribution of the transitory productivity shocks to variance decomposition is the sum of TFP shocks in the final production good (a_t), the one in the domestic investment sector ($a_{d,t}$), and the covariance between them, which absorbs the greater share of the explained variance. In the appendix, we present the disaggregated contribution of each shock.

and of the trade-balance ratio and helps reproduce the observed comovement and volatility ranking.

Table 5: Variance decomposition (%)

Shock	g_y	g_c	g_{i_d}	g_{i_i}	tb_y	rb_y
Production						
Transitory ($a_t, a_{d,t}, \epsilon^{a,ad}$)	80.0	58.1	72.2	3.3	28.6	19.6
Trend (g_t)	15.2	10.9	10.5	0.9	39.8	44.8
Financial						
Spread (μ_t)	2.8	8.1	4.9	2.3	13.2	31.5
Risk (σ_t^d, σ_t^i)	0.6	0.2	0.3	69.5	0.5	0.1
Risk free ($R_{f,t}$)	1.2	3.9	2.6	0.4	5.9	1.8
Other						
Gov. spending (s_t)	0.0	0.2	0.0	0.0	3.0	0.4
Preference (ν_t)	0.1	9.8	0.1	0.0	8.9	1.6
Investment price ($p_{i,t}$)	0.0	0.0	0.0	23.5	0.1	0.1
Measurement error	0.0	8.8	9.4	0.0	0.0	0.2

These variance decompositions might mask important interaction effects between shocks and the endogenous growth mechanism. To isolate this channel, Table 6 compares the second-order moments of our baseline model with those of a counterfactual without endogenous growth. We recalibrate σ_g in the counterfactual so that the volatility of trend growth is unchanged. The standard deviations of the observables then stay nearly constant, so the comparison isolates the change in comovement.

Endogenous growth amplifies the comovement of consumption, output, and investment growth with the trade-balance-to-output ratio. Without endogenous growth, the negative correlation between output growth and the trade balance is 50 percent smaller. Because this strong negative comovement is a defining feature of Sudden Stops, the endogenous trend deepens exactly the joint dynamics that characterize these episodes: the cycle feeds the trend, and the trend in turn shapes the cycle. This two-way feedback between cycle and trend is a distinguishing feature of the emerging-economy business cycle.

Table 6: Second-order moments: baseline vs. no endogenous growth ($\eta = 1$)

	g_y	g_c	g_{i_d}	g_{i_i}	tb_y	rb_y
Correlation with g_y						
Baseline	-	0.91	0.94	0.23	-0.18	-0.10
$\eta = 1$	-	0.90	0.94	0.23	-0.09	-0.08
Correlation with tb_y						
Baseline	-0.18	-0.26	-0.16	-0.07	-	0.71
$\eta = 1$	-0.09	-0.16	-0.10	-0.05	-	0.68

Note: Moments implied by the model at its posterior mean (Baseline, $\eta = 0.69$) and by the counterfactual economy without endogenous growth ($\eta = 1$, with all other parameters at their posterior means and σ_g recalibrated to 0.0205 so that the volatility of trend growth is the same in both economies). Diagonal correlations equal to one are omitted.

The findings of this section suggest that analyzing the contribution of different shocks provides only a partial picture. The full effect of shocks, particularly financial ones, depends crucially on how they interact with the economy's growth process. This fact motivates our examination of financial frictions as key transmission mechanisms, which we explore next.

6.2 The interaction between the trend and financial factors

This section studies whether financial factors affect the trend, one of the key questions we address in this paper. Figure 6 plots the trend response, in percentage deviations from the steady-state trend growth, to a one standard deviation increase in each financial factor.¹⁸

The blue solid line, under the baseline model, shows that a financial shock drives the trend persistently below its steady-state path, so the channel from financial conditions to long-run growth is present and, conditional on a shock, long-lived. All financial shocks generate the same persistent pattern, but their size differs: the spread shock has the largest effect on the trend, the world risk-free rate an intermediate one, and the idiosyncratic-risk shocks the smallest. This is the response conditional on a

¹⁸The steady state trend is defined as $X_{ss,t} = g_{ss} \times t$, where we normalize the initial value of X to 1.

shock. Unconditionally, financial shocks account for only a small share of trend and output fluctuations (Table 5), which are dominated by productivity. What makes financial factors quantitatively important is therefore not the size of financial shocks but the financial frictions that amplify the transmission of the dominant productivity shocks to the trend.

In all cases, the effects are persistent. The transmission mechanism is as follows. An increase in firms' productivity dispersion raises the cost of funding, negatively affecting both imported and domestic investment, leading to a decline in output. This negative effect on capital influences the trend, amplifying the dynamics.

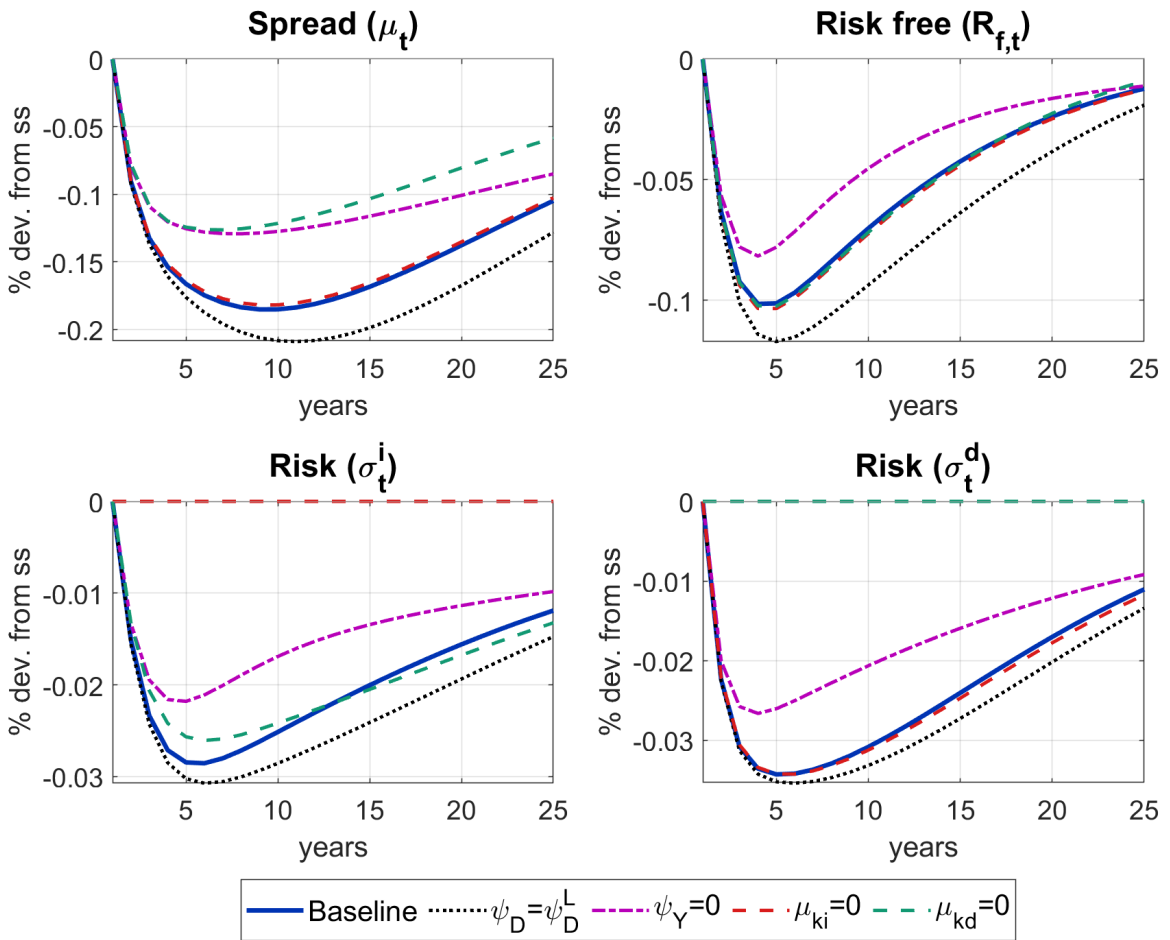


Figure 6: Impulse Response Function (in %) for the trend

Note: Trend (X_t) impulse response function (in logarithms) as percentage deviations (in %) from the steady-state trend growth ($X_{ss,t} = g_{ss} \times t$), to a one standard deviation shock in μ_t , $R_{f,t}$, σ_t^i , and σ_t^d . The trend is obtained as $\ln(X_t) = \ln(X_{t-1}) + \ln(gx_t)$.

The figure also presents a sensitivity analysis under different financial frictions. Notice that the trend response to all shocks is stronger when the debt elastic interest rate coefficient (ψ_D) is set equal to $\psi_D^L = 0.041$, a value 50 percent smaller than its posterior mean. In that case, all shocks also last longer. Here, as firms reduce investment, they also reduce their leverage. The cost of funding, however, does not fall with low ψ_D as much as with the baseline ψ_D , leading to more adjustment in investment.¹⁹

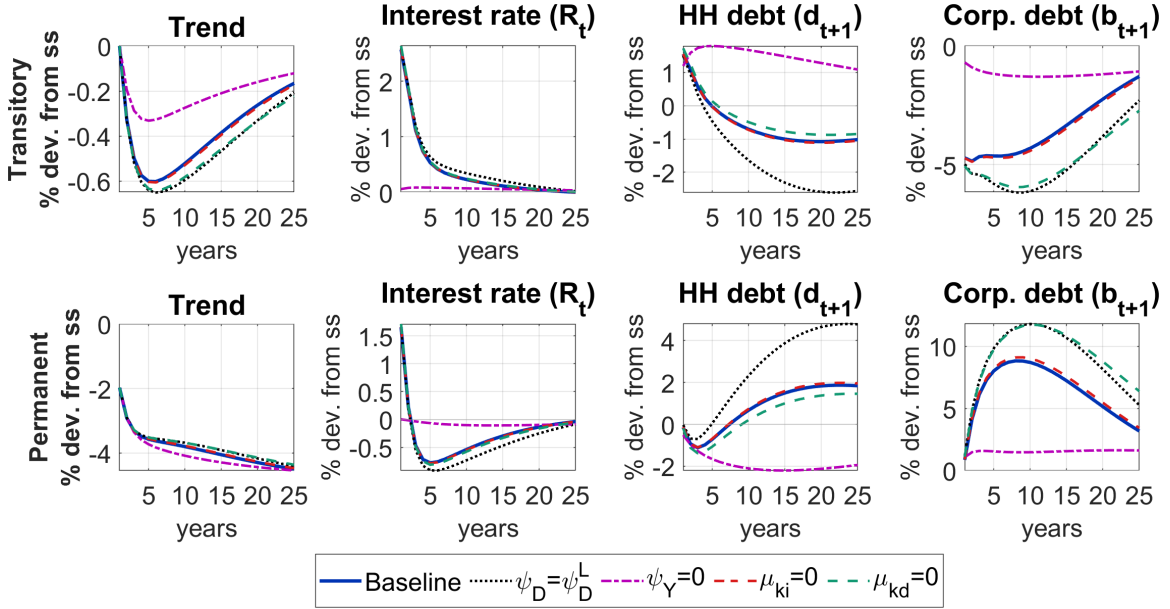


Figure 7: Impulse Response Function (in %) to negative productivity shocks

Note: Trend, interest rate (R_t), household debt (d_{t+1}), and corporate debt (b_{t+1}) impulse response function, as percentage deviations (in %) from the steady state, to a negative one standard deviation shock in $\epsilon_t^{a,ad}$ and g_t .

The screening cost in the imported capital sector also matters. The risk shock materializes through the corporate debt channel: the larger the risk is, the larger is the cost of funds driving down investment incentives. The output elasticity of the interest rate (ψ_Y) has the opposite effect to that of ψ_D . A nonzero ψ_Y slightly amplifies the effect of financial shocks through the interest rate and then on the trend.

¹⁹There is yet some heterogeneity in the responses across investment sectors: for instance, risk and spread shocks with the baseline estimation of μ_{ki} have virtually the same effect on imported investment with ψ_D and ψ_D^L . For completeness, in Section B.6 of the Appendix, we include the impulse responses of several relevant variables for the dynamics discussed in this section.

6.3 The trend and the cycle

Transitory technology shocks play a major role in the variance decomposition. To complete the characterization of the dynamics of the model, we study the effect of productivity shocks and how the transmission channels operate.

Figure 7 shows the response of the interest rate, debt, and the endogenous growth rate of the economy to a negative mean-reverting productivity shock (in the upper block of the figure) and a trend shock (in the lower block). Under the baseline calibration, both shocks cause the endogenous trend to fall below its steady-state level and remain persistently depressed for 25 years, indicating long-lasting effects on growth. The interest rate response differs significantly between the two cases and under different financial friction assumptions because of the different dynamics followed by debt accumulation at the household and corporate levels.

Consider first the baseline model. When the productivity shock is mean-reverting and affects TFP in the final goods production and domestic investment goods production sectors, the interest rate increases. Output falls because technology worsens, labor does not react to wealth effects, and both domestic capital and imported investment capital are fixed for that period. This development induces negative output growth. Households borrow more to smooth consumption, but firms borrow less because they plan to invest less. Overall, the effect is an increase in the interest rate because negative output growth dominates the interest rate dynamics. After that period, output starts recovering, which pushes interest rates down. In the subsequent periods, households' debt decreases, following the return of the interest rate to its steady-state value.

If we decrease the financial friction due to the debt elasticity of the interest rate, the behavior of the interest rate mimics the behavior of the growth rate of output with the opposite sign, as seen by the dotted black line. That said, if we shut down the interest rate elasticity to output growth, the dashed purple line exhibits the hump-shaped behavior of total debt without the initial spike of the interest rate.

When the productivity shock is permanent, the increase in the interest rate is smaller, and from period 1, it falls and remains below its steady-state value for around 20 periods. The role of household debt is critical here, as, given that the economy is permanently poorer, the household does not have incentives to borrow to smooth consumption. Thus, households' debt decreases. This effect generates negative pressure on the interest rate, dominated by the fall in detrended output. Thus, the interest

rate increases in period 0. The response of corporate debt is very small on impact. As the growth rate of the trend returns to its steady state, the interest rate starts falling. Since financing costs go down, corporate debt and then total debt increase, generating the hump shape in the interest rate. Again, if ψ_D is set equal to ψ_D^L , the behavior of R_t mimics the detrended output but with a positive sign, while if we shut down ψ_Y , the behavior of R_t mimics the total debt.

In sum, the previous figures suggest that trend dynamics strongly depend on the cost of financial frictions summarized by the interest rate dynamics. The findings in this section represent one of the main results of the paper. The dichotomy of trend shocks versus financial frictions to explain the business cycle in emerging countries represents a strong simplification, given that the behavior of both of them is closely interconnected.

Figure 8 further explores the role of capital by showing how domestic and imported capital transmit each shock to the trend. The figure plots the response of the endogenous trend to each structural shock in three economies: the baseline (in solid blue), one in which imported capital is fixed to its steady state value in the trend equation (with green triangles), and one in which capital is removed from the trend altogether (in red dashed lines). Two features stand out. First, the figure isolates the endogenous-growth mechanism of the model: the trend responds directly only to the permanent productivity shock, while every other shock reaches it exclusively through capital, so the “capital removed” line is flat at zero. Second, the shocks that are specific to the external margin, namely risk in the imported-capital sector, the imported-investment price, and the spread shock, to a lesser extent, transmit to the trend mainly through imported capital, and these are precisely the shocks that dominate the imported-investment and trade-balance dynamics of Sudden Stops. Shocks that are not specific to that margin, such as transitory productivity and domestic-sector risk, reach the trend mostly through domestic capital. Imported capital therefore matters most on the external margin, through which financial and external conditions leave a persistent scar on long-run growth.

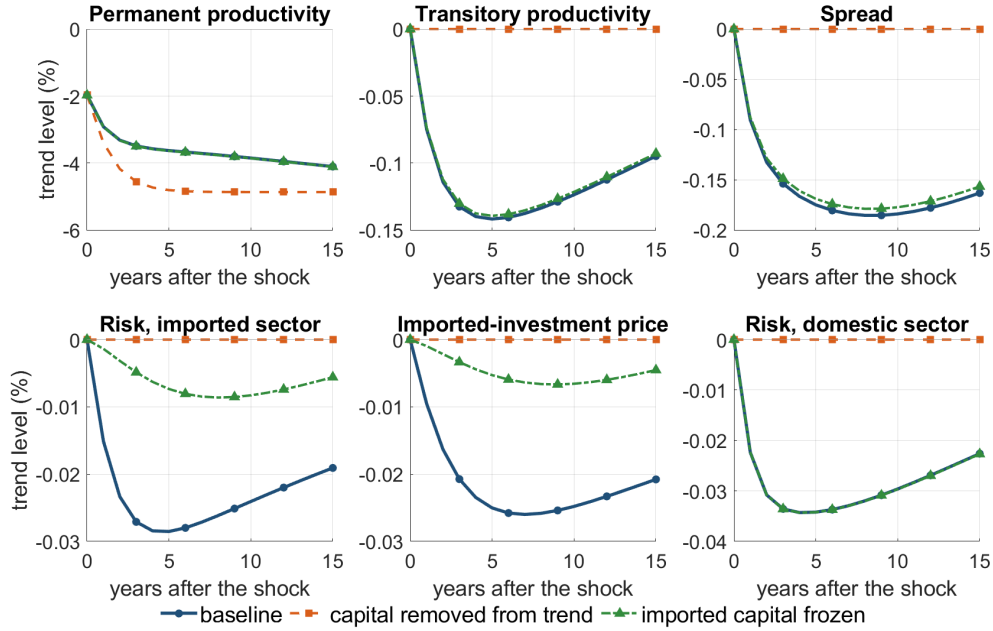


Figure 8: Which capital transmits each shock: impulse response of the endogenous trend

Note: Cumulative response of the endogenous trend X_t (percent) to a one-standard-deviation adverse structural shock, negative for the productivity shocks, so every panel shows a trend decline, at the posterior means. Solid: baseline. Dashed: capital removed from the trend equation (16), with η held fixed. Dash-dotted: imported capital frozen at its steady-state value in equation (16). The permanent productivity shock moves the trend directly; every other shock reaches the trend only through capital, so the “capital removed” line is flat at zero. Imported-sector shocks (risk in the imported sector, imported-investment price) are transmitted mainly by imported capital, freezing it removes most of the response; domestic and aggregate shocks (transitory productivity, spread, domestic-sector risk) are transmitted by domestic capital, freezing imported capital leaves the response unchanged.

7 The long-run effect of Sudden Stops

Sudden Stops are the most common financial crisis in emerging countries: they combine an increase in spreads, capital flow reversals, and an output fall. In this section, we study the effect of Sudden Stops on the long-run growth of the economy. We do it in two steps. First, we show that the model can replicate the dynamics around Sudden Stops. Second, we study the permanent effect Argentina’s trend suffered around each Sudden Stop identified in the data. This question becomes relevant to under-

standing the long-run implications of short-run macroeconomic volatility in emerging countries.

7.1 Anatomy of Sudden Stops

This section studies the model implications for the average simulated Sudden Stop, the percentiles 32 and 68, and compares it with the average in the Argentine data. As with the data, we define a Sudden Stop episode as a year in which the country presents a 2 percent fall in the GDP and a 2 percentage point increase in the net-exports-to-output ratio. We simulate the economy for 500,000 periods, remove the first half of the observations, recover the episodes that fit into our definition of a Sudden Stop, and compute the cross-sectional average of all episodes, including the five years before and after. As in the data, when two episodes are fewer than five years apart we keep only the first, measuring the spacing against the last retained episode. Applying this same rule to the simulation, the frequency of Sudden Stops implied by the model, 7.9 percent, is of comparable magnitude to the 9.4 percent observed in the data, a moment the estimation does not target.

Figure 9 shows that the model reproduces the main dynamics of a Sudden Stop. The output collapse is matched almost exactly, about 6 percent at impact in both the data and the model, and the direction, timing, and persistence of consumption, domestic investment, the trade balance, and interest payments are all present. The average simulated episode is milder than the realized Argentine ones on the imported-investment growth, but the cross-episode dispersion is large, with a one-standard-deviation band of roughly 45 percentage points, so the model’s average lies well within it. This gap is not a sign of an over-smooth model, which reproduces the very high unconditional volatility of imported investment (Table 4, 45.9 against 41.3 percent). The 2002 crisis, the deepest episode in the sample, accounts for a sizable part of this gap: excluding it brings the average fall of imported investment in the data closer to the model, though the model remains somewhat milder (Appendix Figure 18).²⁰ This works, if anything, in our favor: even the milder collapse of imported investment that the model produces is enough to generate a persistent effect on the trend, so the channel we quantify is a conservative estimate of its strength.

²⁰We highlight the importance of 2002 here because that crisis was a sovereign default crisis, a feature that our model abstracts from.

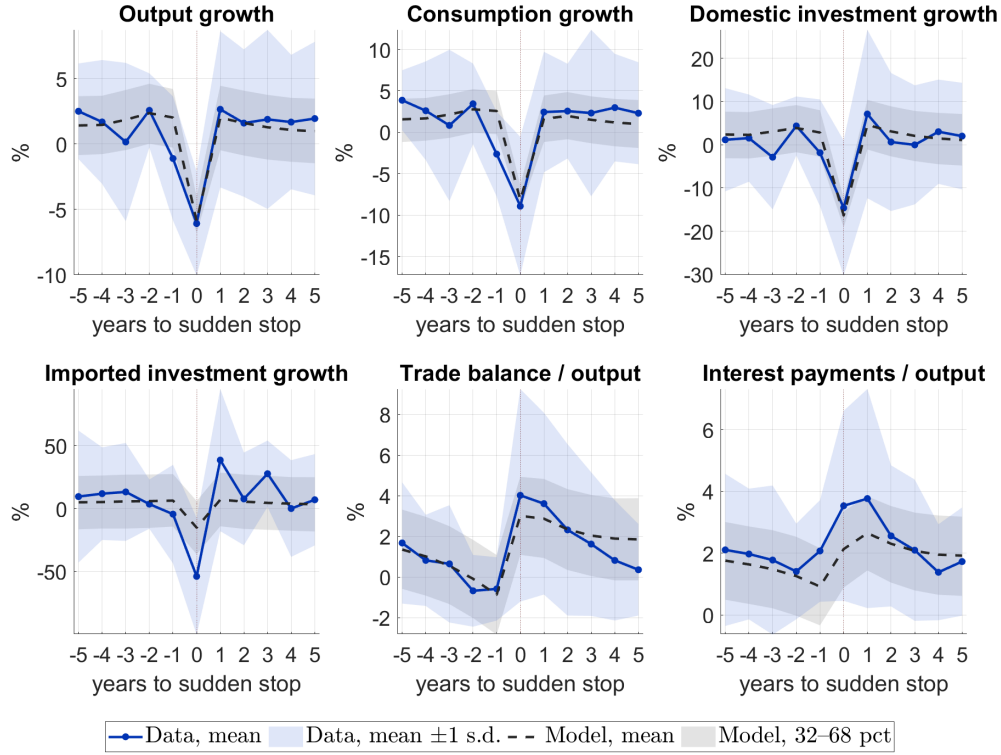


Figure 9: Model versus data around Sudden Stops

Note: Argentine data event study (solid line with markers: cross-episode mean; light band: mean $\pm$ one standard deviation across the six Sudden Stops of 1959, 1976, 1982, 1988, 1995 and 2002) against the average simulated Sudden Stop in the estimated model (dashed line; shaded band: 32nd–68th percentiles across simulated episodes), for output growth, consumption growth, domestic and imported investment growth, the trade-balance-to-output ratio, and interest payments to output. The 2002 episode is the largest in the data.

In the following section, we show the Sudden Stop emerges endogenously as a combination of various technological and financial shocks that affect the economy differently depending on the episode.

7.2 A quantitative view of the long-run effect of Sudden Stops

We now turn to quantifying the effects of Sudden Stops on trend growth, episode by episode. Figure 10 presents the historical decomposition of the trend around each Sudden Stop for the baseline model, and the counterfactuals fixing imported and total

capital in the trend. We group the shocks in “productivity shocks”, “financial shocks” and “other shocks”.

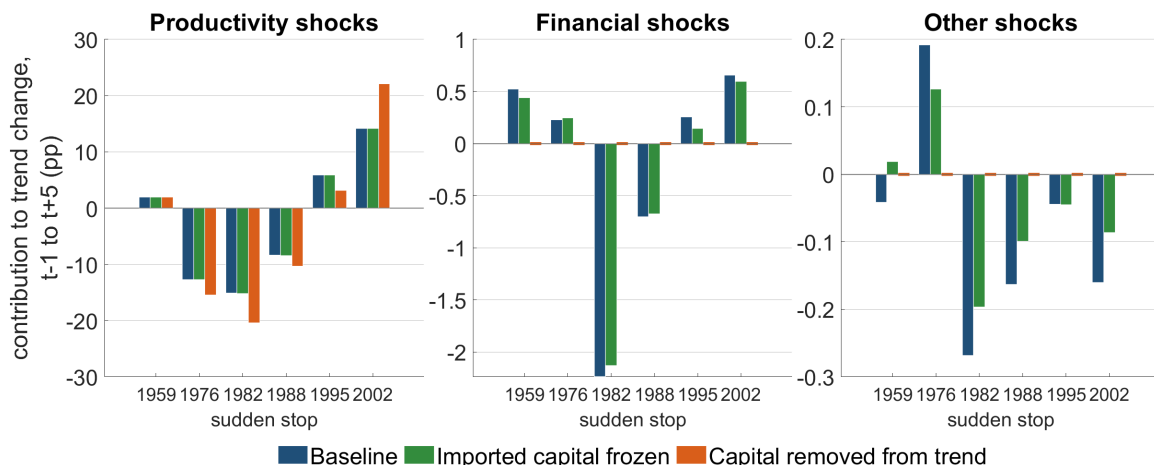


Figure 10: Historical decomposition of the endogenous trend for the three economies

Note: Contribution (percentage points) of each shock block to the change in the endogenous trend from $t - 1$ to $t + 5$, by Argentine Sudden Stop, feeding the estimated smoothed shocks through each economy. A block’s contribution is the trend change with all shocks minus the trend change with that block shut down. Bars: baseline (blue), imported capital frozen in Eq. (16) (green), capital removed from Eq. (16) (orange).

Figure 10 reports, for each of the six Argentine Sudden Stops, the contribution of the productivity, financial, and other shock blocks to the change in the endogenous trend, in the three economies. Three points are worth highlighting. First, by construction, in the economy with capital removed from the trend equation (where the trend becomes fully exogenous), the financial and other shocks contribute exactly zero in every episode (the orange ticks at the axis), so the endogenous-capital term is the only channel through which financial and transitory disturbances reach long-run growth. The productivity block, which also has a direct exogenous-trend component, remains in all three economies. Second, freezing only imported capital leaves the historical contribution of productivity shocks close to the baseline in every episode: at the realized episodes the level of the trend is dominated by productivity, which propagates through capital of both types, so the specific role of imported capital for the level is small and is instead isolated shock by shock in the impulse responses of Figure 8, where the external shocks that dominate the imported-investment margin of

Sudden Stops travel through imported capital. Nevertheless, imported capital plays a non-negligible role in transmitting financial shocks, and a sizable role for other shocks, as can be observed in the third block of the figure for all episodes except 1995.

Finally, Figure 11 separates the exogenous and endogenous components of the trend around each of the six episodes. For every Sudden Stop it plots the smoothed trend against counterfactual paths in which one shock block at a time is shut down, each computed under the baseline weight of the endogenous component ($\eta = 0.69$), a fully exogenous trend ($\eta = 1$), and a fully endogenous trend ($\eta = 0$). The persistence of the trend response comes from the endogenous component: with a fully exogenous trend the counterfactual reverts quickly toward the smoothed path, whereas the endogenous component turns the crisis into a drawn-out decline in the trend. This is the central mechanism of the paper, through which the business cycle translates into persistent long-run stagnation.

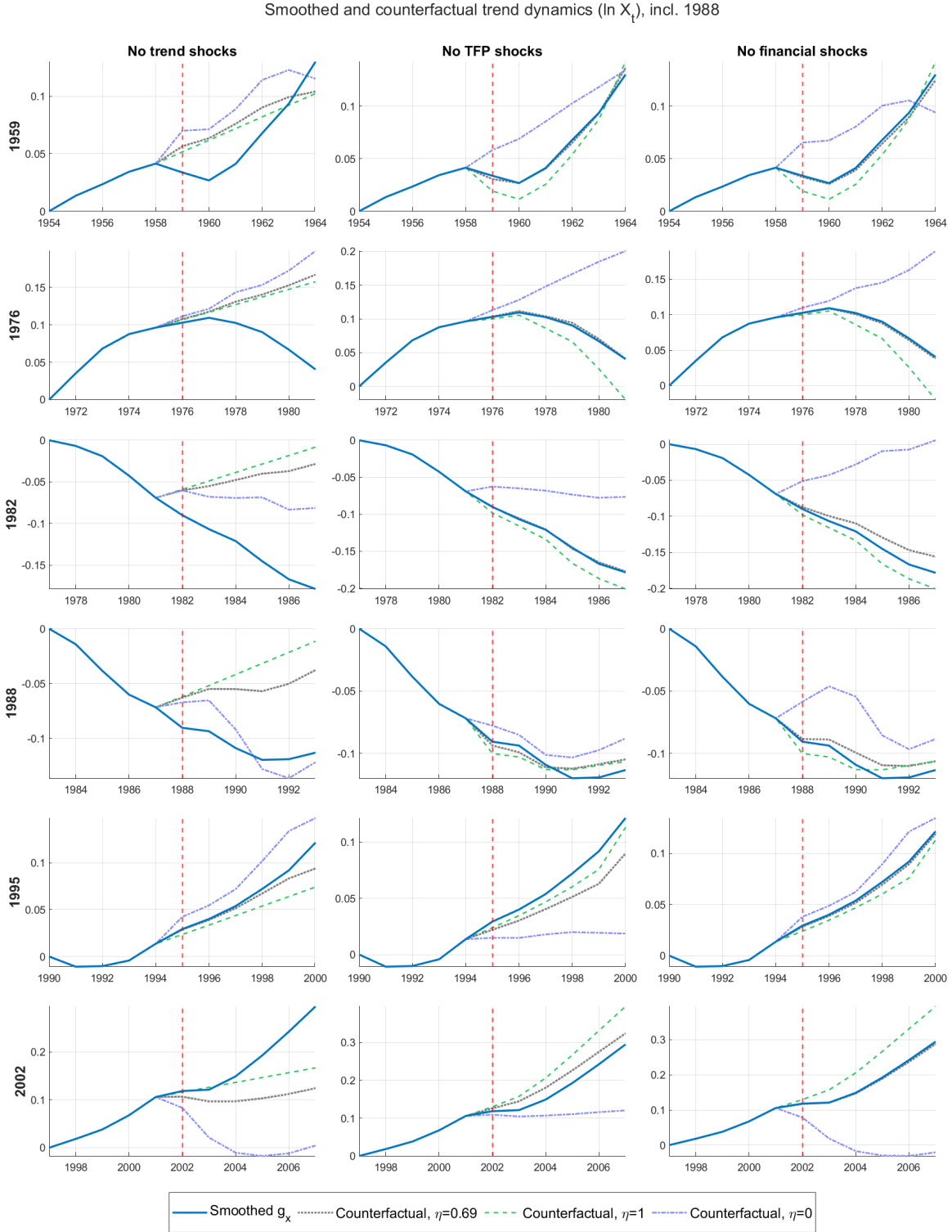


Figure 11: Smoothed and counterfactual trend dynamics

Note: For each of the six Argentine Sudden Stops (rows), the log endogenous trend $\ln X_t$ around the crisis year. Columns shut down one shock block at a time: trend shocks, TFP shocks, and financial shocks. In each panel the smoothed trend (solid blue) is compared with counterfactual paths under the baseline endogenous-component weight ($\eta = 0.69$), a fully exogenous trend ($\eta = 1$), and a fully endogenous trend ($\eta = 0$).

8 Concluding remarks

We develop and estimate a framework that links Sudden Stops, imported-capital accumulation, and endogenous growth to explain why emerging economies combine low long-run growth, high volatility, and frequent Sudden Stops. In the model, disruptions to investment during a Sudden Stop leave a persistent mark on long-run growth. Estimated on sixty-five years of Argentine data, the framework lets us measure the forces behind the trend.

A central innovation of our framework is that it endogenizes the trend through capital accumulation. [Aguiar and Gopinath \(2007\)](#) argue that the trend volatility is key for the business cycle of emerging economies. [Garcia-Cicco et al. \(2010\)](#), in contrast, highlight the role of financial frictions. Our theory brings the two together, capturing the interaction between trend volatility and financial frictions in a quantitative real model of the business cycle and the long-run behavior of Argentina. The evolution of the trend is not purely exogenous, but also reflects financial conditions and investment dynamics. Quantitatively, the movements of the trend are driven by productivity, while financial factors matter not as a direct source of trend fluctuations but as the frictions and the imported-capital channel through which those shocks are transmitted and made persistent.

Our analysis yields three main insights for emerging-market dynamics. First, while the literature has debated the relative importance of trend versus transitory shocks, we show that the two are interconnected: because the trend responds to capital accumulation, even a transitory shock acquires lasting effects. The historical decomposition of the Argentine episodes illustrates this. During the 1982 crisis, financial constraints and risk spreads shaped not only the immediate trade and consumption dynamics but also, through investment, the subsequent path of the trend.

Second, our findings highlight the dual role of financial factors in emerging markets, both as sources of shocks and as transmission mechanisms. Financial frictions prove particularly important for imported capital, with a monitoring cost about an order of magnitude larger than in the domestic sector. This asymmetry creates a powerful channel through which financial shocks reach the economy beyond their immediate impact: when conditions tighten, the higher friction in the imported-capital sector magnifies the shock and propagates it to longer horizons through the endogenous trend.

Third, endogenous growth amplifies crisis dynamics. Without it, the correlation between output growth and the trade balance would be about fifty percent weaker, so the feedback between the cycle and the trend deepens the joint movements that define a Sudden Stop. Models that abstract from this linkage may substantially understate the cost of emerging-market crises.

Finally, our results point to two directions for future research. First, the externality that runs from capital accumulation to the trend suggests a role for policies acting on imported capital, such as capital controls on foreign borrowing or tariffs on imported capital goods. These instruments shift the level of the balanced-growth path rather than pre-empt crises, and a tariff on capital goods can be counterproductive in the medium run. Because a credible welfare ranking would require a higher-order solution and the transition path, we leave a formal optimal-policy analysis to future work. Second, channels other than capital accumulation, such as human capital, technology adoption, or firm dynamics, likely also link crises to growth, and quantifying their relative importance across countries could yield further insights.

9 Bibliography

- AGUIAR, M. AND G. GOPINATH (2006): “Defaultable debt, interest rates and the current account,” Journal of international Economics, 69, 64–83.
- (2007): “Emerging market business cycles: The cycle is the trend,” Journal of political Economy, 115, 69–102.
- AKINCI, Ö. (2021): “Financial Frictions and Macro-Economic Fluctuations in Emerging Economies,” Journal of Money, Credit and Banking, 53, 1267–1312.
- ALESSANDRIA, G., S. PRATAP, AND V. YUE (2015): “Export growth and aggregate dynamics in large devaluations,” Tech. rep., mimeo.
- ANZOATEGUI, D., D. COMIN, M. GERTLER, AND J. MARTINEZ (2019): “Endogenous technology adoption and R&D as sources of business cycle persistence,” American Economic Journal: Macroeconomics, 11, 67–110.
- ATES, S. T. AND F. E. SAFFIE (2021): “Fewer but better: Sudden stops, firm entry, and financial selection,” American Economic Journal: Macroeconomics, 13, 304–356.
- BENGURIA, F., H. MATSUMOTO, AND F. SAFFIE (2022): “Productivity and trade dynamics in sudden stops,” Journal of International Economics, 139, 103631.
- BERNANKE, B. S., M. GERTLER, AND S. GILCHRIST (1999): “The financial accelerator in a quantitative business cycle framework,” Handbook of macroeconomics, 1, 1341–1393.
- BIANCHI, F., H. KUNG, AND G. MORALES (2019): “Growth, slowdowns, and recoveries,” Journal of Monetary Economics, 101, 47–63.
- BONCIANI, D., D. GAUTHIER, AND D. KANNGIESSER (2023): “Slow Recoveries, Endogenous Growth and Macro-prudential Policy,” Review of Economic Dynamics, 51, 698–715.
- CALVO, G. A., A. IZQUIERDO, AND E. TALVI (2006): “Sudden stops and phoenix miracles in emerging markets,” American Economic Review, 96, 405–410.

- CHANG, R. AND A. FERNÁNDEZ (2013): “On the sources of aggregate fluctuations in emerging economies,” International Economic Review, 54, 1265–1293.
- CHRISTIANO, L. J., R. MOTTO, AND M. ROSTAGNO (2010): “Financial Factors in Economic Fluctuations,” ECB Working Paper 1192, European Central Bank.
- (2014): “Risk shocks,” American Economic Review, 104, 27–65.
- COMIN, D. (2004): “R&D: A small contribution to productivity growth,” Journal of Economic Growth, 9, 391–421.
- COMIN, D. AND M. GERTLER (2006): “Medium-term business cycles,” American Economic Review, 96, 523–551.
- CUGAT, G. AND A. MANERA (2024): “Do Capital Inflows Spur Technology Diffusion? Evidence from a New Technology Adoption Index,” IMF Working Paper 2024/044, International Monetary Fund.
- FERNÁNDEZ, A. AND A. GULAN (2015): “Interest rates, leverage, and business cycles in emerging economies: The role of financial frictions,” American Economic Journal: Macroeconomics, 7, 153–188.
- FERNÁNDEZ-VILLAYERDE, J. (2010): “Fiscal Policy in a Model with Financial Frictions,” American Economic Review: Papers and Proceedings, 100, 35–40.
- GARCIA-CICCO, J., R. PANCRAZI, AND M. URIBE (2010): “Real business cycles in emerging countries?” American Economic Review, 100, 2510–31.
- GAULIER, G. AND S. ZIGNAGO (2010): “BACI: International Trade Database at the Product-Level. The 1994-2007 Version,” Working Paper 2010-23, CEPIL.
- GOPINATH, G. AND B. NEIMAN (2014): “Trade adjustment and productivity in large crises,” American Economic Review, 104, 793–831.
- GREENWOOD, J., Z. HERCOWITZ, AND G. W. HUFFMAN (1988): “Investment, capacity utilization, and the real business cycle,” The American Economic Review, 78, 402–417.
- GUERRON-QUINTANA, P. A. AND R. JINNAI (2019): “Financial frictions, trends, and the great recession,” Quantitative Economics, 10, 735–773.

- INSTITUTO INTERDISCIPLINARIO DE ECONOMÍA POLÍTICA DE BUENOS AIRES (IIEP, UBA-CONICET) (2018): “Base de datos e indicadores macro para la economía argentina,” Universidad de Buenos Aires – CONICET, long historical series on Argentine national accounts. Accessed 2018.
- JORDÀ, Ò., K. KNOLL, D. KUVSHINOV, M. SCHULARICK, AND A. M. TAYLOR (2019): “The rate of return on everything, 1870–2015,” The Quarterly Journal of Economics, 134, 1225–1298.
- JUSTINIANO, A. AND B. PRESTON (2010): “Monetary policy and uncertainty in an empirical small open-economy model,” Journal of Applied Econometrics, 25, 93–128.
- MATSUMOTO, H. (2022): “Foreign reserve accumulation, foreign direct investment, and economic growth,” Review of Economic Dynamics, 43, 241–262.
- MENDOZA, E. G. AND V. Z. YUE (2012): “A general equilibrium model of sovereign default and business cycles,” The Quarterly Journal of Economics, 127, 889–946.
- MIYAMOTO, W. AND T. L. NGUYEN (2017): “Business cycles in small open economies: Evidence from panel data between 1900 and 2013,” International Economic Review, 58, 1007–1044.
- NEUMEYER, P. A. AND F. PERRI (2005): “Business cycles in emerging economies: the role of interest rates,” Journal of monetary Economics, 52, 345–380.
- OTTONELLO, P. AND T. WINBERRY (2024): “Capital, Ideas, and the Costs of Financial Frictions,” NBER Working Paper No. 32056.
- PARK, J. (2017): “Sovereign default and capital accumulation,” Journal of International Economics, 106, 119–133.
- QUERALTO, A. (2020): “A model of slow recoveries from financial crises,” Journal of Monetary Economics, 114, 1–25.
- SEOANE, H. D. (2016): “Parameter drifts, misspecification and the real exchange rate in emerging countries,” Journal of International Economics, 98, 204–215.
- SEOANE, H. D. AND E. YURDAGUL (2019): “Trend shocks and sudden stops,” Journal of International Economics, 121, 103252.

URIBE, M. AND S. SCHMITT-GROHÉ (2017): Open economy macroeconomics, Princeton University Press.

URIBE, M. AND V. Z. YUE (2006): “Country spreads and emerging countries: Who drives whom?” Journal of international Economics, 69, 6–36.

A Online Appendix

A.1 Data sources and construction

This appendix documents the source, classification, and construction of every variable that enters the empirical analysis in Section 2 and the estimation in Section 4. The cross-country facts draw on three inputs: macroeconomic series in constant local currency from the World Bank World Development Indicators (WDI), a within-country spliced panel of merchandise imports by end-use category built from UN Comtrade and CEPII's BACI database, and additional World Bank indicators for external-debt interest and for research and development ($R\&D$). The Argentine estimation draws on the long historical national accounts of the *Instituto Interdisciplinario de Economía Política de Buenos Aires* (Instituto Interdisciplinario de Economía Política de Buenos Aires (IIEP, UBA-CONICET), 2018). Table 7 summarizes the sources and the subsections below describe each variable group in detail.

A.1.1 Country sample and classification

We follow the classification of Uribe and Schmitt-Grohé (2017). A country is classified as emerging if its average PPP-converted GDP per capita over 1990–2017 lies between 3,000 and 25,000 US dollars of 2005, and as advanced if it lies above that range. Within the emerging group we retain small open economies from Latin America and the Caribbean, Europe and Central Asia, and the Middle East and North Africa. For the advanced comparison group we use the developed small open economies of Garcia-Cicco et al. (2010). The resulting lists are reported in the note to Table 1: 34 emerging small open economies (of which 11 are Latin American) and 14 developed small open economies. The income and PPP classification is assigned once per country from these criteria and applied consistently across all cross-country figures and tables.

A.1.2 Sudden Stop episodes

We define a Sudden Stop, following Seoane and Yurdagul (2019), as a year in which real GDP per capita falls by at least 2 percent and the trade-balance-to-output ratio rises by at least 2 percentage points. Formally, an episode is a year t with $\Delta \ln y_t \leq -0.02$ and $\Delta(tb/y)_t \geq 0.02$, where y is real GDP per capita (Section A.1.3) and tb/y is the trade-balance-to-output ratio.

Two additional rules define the event-study sample. First, a 5-year spacing (“retained”) rule: when two candidate years occur within five years of each other, the later candidate is retained only if more than five years separate it from the previous retained episode, so that no two retained episodes lie within five years. Second, an episode is discarded if its symmetric five-year event window $[t - 5, t + 5]$ falls outside the sample. Applied to the emerging sample this procedure yields 59 Sudden Stop episodes. Because individual variables have different coverage, each panel of Figure 1 uses every episode for which the variable is observed over the window: output growth and net exports are computed on the full set of 59 episodes, export and import growth on 50, imported-capital growth on 43 (1976 onward), and interest payments on 37. The category decomposition in Figure 2 uses the 37 episodes with full-window composition coverage, contrasted with 128 ordinary-downturn years (years of negative real GDP growth outside Sudden Stops, in 23 countries).

A.1.3 Output

Real output is real GDP per capita in constant local currency units. The series is the World Bank World Development Indicators (WDI) indicator `NY.GDP.PCAP.KN` (“GDP per capita, constant LCU”). Because the series is already expressed per capita, output growth is the annual log change of the level, and it coincides by construction with WDI real GDP per-capita growth.

A.1.4 Trade balance, exports, and imports

The trade-balance-to-output ratio (tb/y) is constructed as nominal exports minus nominal imports over nominal output, all in local currency, from the World Bank national accounts. Its year-over-year change is the second condition in the Sudden Stop definition and is plotted as the net-exports panel of Figure 1. The exports-to-output and imports-to-output ratios (X/Y , M/Y), used in the net-exports decomposition of Appendix A.3, are formed analogously from the same nominal series.

Export and import growth are annual log changes of the real per-capita series. Real exports and imports are divided by population to obtain per-capita volumes, and growth is the log change of those per-capita series.

A.1.5 Population

Population is total population from the World Bank (WDI indicator `SP.POP.TOTL`), complemented where necessary by the Federal Reserve Economic Data (FRED).

A.1.6 Imported capital goods

Imported capital goods are measured throughout the paper by a single within-country spliced series. It measures imports of capital goods, defined as the sum of two Broad Economic Categories (BEC) end-use classes: capital goods except transport equipment (BEC 41) and industrial transport equipment (BEC 521). We describe the classification framework, then the construction of the series, then its coverage relative to an alternative source.

UN Comtrade, BACI, and the BEC classification. All merchandise trade data originate from UN Comtrade reporter declarations. The BEC classification partitions traded goods by whether their principal use is as capital, intermediate, or consumption goods. The classes BEC 41 (capital goods, excluding transport equipment) and BEC 521 (transport equipment, industrial) together define capital goods. BACI is the CEPII harmonization of Comtrade bilateral flows ([Gaulier and Zignago, 2010](#)): it reconciles the mirror (importer and exporter) declarations of each flow into a single cleaned value at the Harmonized System (HS) six-digit level. Capital goods are obtained by concordance from HS product codes for BACI and from SITC product codes for the earlier Comtrade vintages.

Construction of the spliced series. The series is built to span the full sample, including the pre-1996 episodes that predate the BACI era, as follows.

1. *BACI-HS, 1996 onward.* CEPII BACI HS96 flows (release V202501, 1996–2023, thousands of current USD) are aggregated to country-year imports by end-use class through a concordance from HS six-digit codes to BEC. Capital is BEC 41 plus BEC 521. The other classes are intermediate and consumption goods.
2. *Comtrade SITC Rev.2, before 1996.* For years before 1996 we take UN Comtrade imports at the SITC (3-digit) level, world partner, from the Rev.2 vintage (1976–1995), and map them to the same BEC buckets through a documented

SITC Rev.2 to BEC concordance. The capital bucket reproduces the BEC 41 plus 521 anchor (machinery groups, office and telecom equipment, professional instruments, and industrial and non-passenger transport equipment such as goods vehicles, railway stock, aircraft, and ships), while passenger cars are assigned to consumption.

3. *Comtrade SITC Rev.1, for the earliest windows.* For 1980s debt-crisis episodes whose windows predate Rev.2 coverage, we add UN Comtrade SITC Rev.1 imports (1962–1990) through an analogous Rev.1 to BEC concordance. Because the Rev.1 road-vehicle category mixes passenger cars and commercial vehicles, we assign to capital only its commercial-vehicle portion, using the commercial-vehicle share of road-vehicle imports implied by the later Rev.2 classification. The Sudden Stop composition dynamics are insensitive to this split (within-country growth correlations above 0.97 whether the category is fully excluded or included).
4. *Overlap gate and level rescaling.* On the 1996–2004 window, where BACI-HS and Comtrade-SITC both exist, we compare the two capital series within country. They comove closely on within-country growth and shares, with a stable per-country level offset. We therefore accept the splice for growth and share objects and, for levels, rescale the pre-1996 Comtrade level onto the BACI scale by the per-country median overlap ratio (a global median ratio where a country has too few overlap years), so that the pre-1996 level is continuous with BACI. An analogous within-country overlap gate certifies the Rev.1 series against Rev.2 before it is spliced in, and non-reconciling countries are excluded from the spliced panel.
5. *Precedence and provenance.* Where more than one source covers a country-year, precedence is BACI-HS, then SITC Rev.2, then SITC Rev.1. The provenance of every observation is recorded, so that each country-year is attributed to a single source.

Coverage and an alternative source. This series is used for every object in the paper that involves imported capital: the imported-capital panel of the event study (Figure 1), the import-category decomposition (Figure 2, capital versus intermediate versus consumption), the R&D-reliance measure (Figure 3), and the recovery scatter

reported in the text. Its pre-1996 depth is essential: it covers the imported-capital collapse of the major early debt-crisis episodes, for example Chile 1982, Brazil 1983, and Uruguay 1983. These are missed by the post-1996 BACI data and by SITC Rev.2 alone, whose reporting for these countries begins only in the mid-1980s, and are recovered here through SITC Rev.1. On the 1996–2004 window where the pre-1996 Comtrade series and BACI overlap, the two track each other closely within country, with a median growth correlation of about 0.9 and a level correlation of 0.98, which is why we splice them for the within-country event studies.

A.1.7 Other import categories

Intermediate and consumption imports, used alongside capital in the category decomposition of Figure 2, come from the same spliced series described above. For BACI-HS years they are the HS six-digit to BEC intermediate and consumption classes. For the pre-1996 Comtrade extensions they are the corresponding SITC Rev.2 and Rev.1 to BEC buckets (intermediate covers crude materials, fuels, intermediate chemicals, manufactured materials, and parts; consumption covers processed food and beverages, household goods, passenger cars, and other final consumer manufactures). Shares are computed against total merchandise imports.

A.1.8 Interest payments

The interest-payments-to-output ratio (rb/y) is interest payments on external debt from the World Bank International Debt Statistics (indicator `DT.INT.DECT.CD`, current US dollars) divided by GDP in current US dollars (WDI indicator `NY.GDP.MKTP.CD`). In the event study of Figure 1 it is shown as the change from the pre-crisis ($t - 1$) level, in percentage points of GDP. This variable also enters the Argentine estimation (see below).

A.1.9 Research and development expenditure

R&D expenditure over GDP is the World Bank WDI indicator `GB.XPD.RSDV.GD.ZS` (gross domestic expenditure on R&D as a percent of GDP). The reliance figure reads the country-level mean of this series over the sample period.

A.1.10 Argentine estimation data

The model is estimated on Argentina over 1951–2015. The national accounts come from the *Instituto Interdisciplinario de Economía Política de Buenos Aires* ([Instituto Interdisciplinario de Economía Política de Buenos Aires \(IIEP, UBA-CONICET\), 2018](#)), which provide long historical series that separately measure domestic and imported investment in transport material and in machinery and equipment. This separation is what allows us to discipline the domestic and imported capital sectors of the model independently. Total imported investment is the sum of imported transport and imported machinery and equipment; domestic investment is the sum of construction, domestic transport, and domestic machinery and equipment.

The estimation uses eight observables, all in per-capita terms and, except for the two ratios, in natural logs: (1) GDP growth (g_y); (2) private consumption growth (g_c); (3) domestic investment growth (g_{i_d}); (4) imported investment growth (g_{i_i}); (5) the trade-balance-to-output ratio (tb_y); (6) the growth rate of the relative price of imported capital investment goods to the GDP deflator (g_{p_i}); (7) the world (U.S.) real risk-free interest rate (R_f); and (8) the ratio of net interest payments on the foreign net asset position to output (rb_y). Output, consumption, and government spending are deflated by the GDP deflator; domestic and imported investment are deflated by their own deflators, so all deflators are Fisher-chained indexes. The relative price of imported investment is the implicit deflator of imported investment (nominal over real imported investment) divided by the GDP deflator. The real risk-free rate is the short-term U.S. nominal rate from [Jordà et al. \(2019\)](#), made real by subtracting current U.S. CPI inflation and then demeaned. The six Argentine Sudden Stops obtained by applying the definition of Section [A.1.2](#) to these data are 1959, 1976, 1982, 1988, 1995, and 2002. Figure [13](#) plots the eight observables together with their smoothed counterparts, and Figure [12](#) shows the relative price of imported investment, which rises around the 1982, 1988, and 2002 crises.

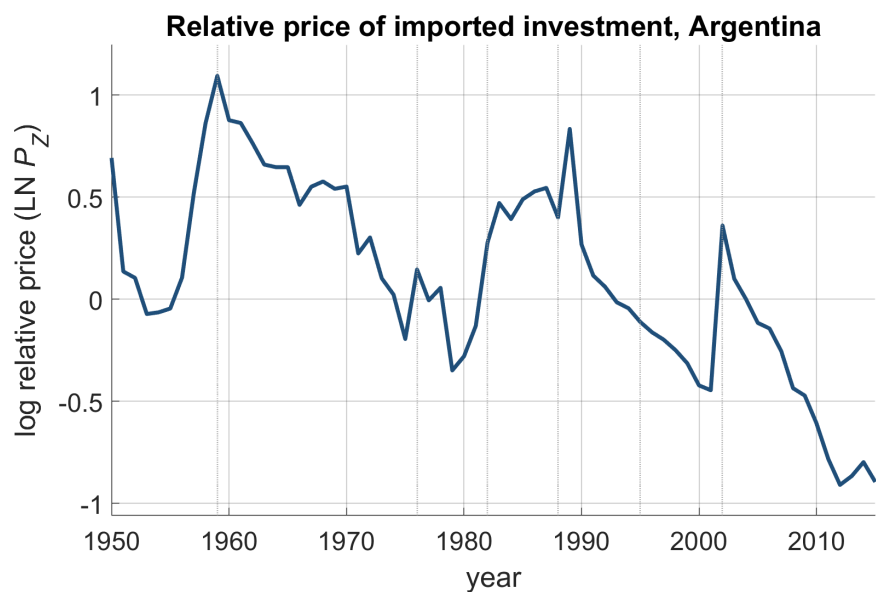


Figure 12: Relative price of imported investment in Argentina

Note: Log relative price of imported investment goods in Argentina (the series $\ln P_Z$ used in the estimation), from [Instituto Interdisciplinario de Economía Política de Buenos Aires \(IIEP, UBA-CONICET\)](#) (2018). Dotted lines mark the Argentine Sudden Stops. The series trends down over the sample and rises around the 1982, 1988, and 2002 crises.

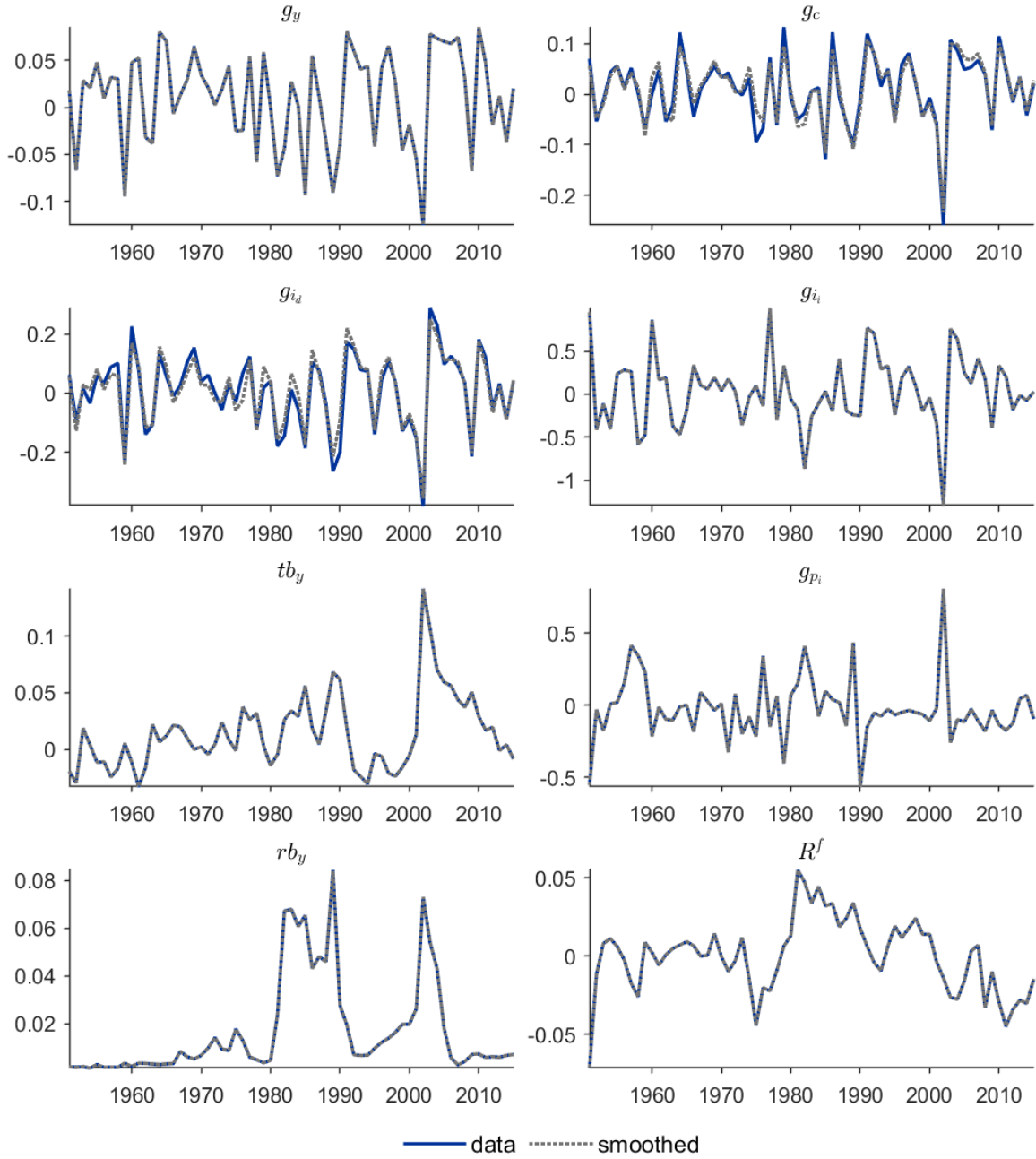


Figure 13: Observable variables and smoothed variables

Time series used in the Bayesian estimation: output growth (g_y); private consumption growth (g_c); domestic investment growth (g_{i_d}); imported investment growth (g_{i_i}); the trade-balance-to-output ratio (tb_y); interest payments (rb_y); the growth rate of the relative price of imported capital investment goods to the GDP deflator, demeaned (g_{p_i}); and the risk-free interest rate (R^f). The blue line is the data; the gray dashed line is the smoothed variable.

Table 7: Data sources and construction: summary

Variable	Source	Coverage (years)	Used in
Real GDP per capita	World Bank WDI (NY.GDP.PCAP.KN); constant LCU, per capita	1960–2018	SS definition; Fig. 1, 5
Trade balance / GDP	World Bank national accounts; nominal $(X - M)/Y$, LCU	1960–2018	SS definition; Fig. 1
Exports / GDP, imports / GDP	World Bank national accounts; nominal ratios, LCU	1960–2018	Appx. A.3
Exports, imports per capita	World Bank national accounts and population; real, per capita (log growth)	1960–2018	Fig. 1, 5; Appx. A.2
Imported capital goods	Spliced CEPII BACI-HS and UN Comtrade SITC Rev.2 / Rev.1 (Gaulier and Zignago, 2010); BEC 41+521, overlap-gate rescaled	1962–2023	Fig. 1, 2, 3; recovery scatter
Intermediate, consumption imports	Spliced BACI-HS and Comtrade SITC; BEC end-use classes	1962–2023	Fig. 2
Interest payments / GDP	World Bank IDS (DT.INT.DECT.CD) over GDP (NY.GDP.MKTP.CD); current USD	1970–2018	Fig. 1; estimation (rb_y)
R&D expenditure / GDP	World Bank WDI (GB.XPD.RSDV.GD.ZS); percent of GDP	country mean	Fig. 3
Population	World Bank WDI (SP.POP.TOTL) and FRED; total population	1960–2018	per-capita conversions
Argentine national accounts	IIEP (Instituto Interdisciplinario de Economía Política de Buenos Aires (IIEP, UBA-CONICET), 2018); domestic vs. imported transport and equipment	1951–2015	estimation; Fig. 5
U.S. risk-free rate	Jordà et al. (2019); short-term nominal, real ex-post (CPI) and demeaned	1951–2015	estimation (R_f)

A.2 Some facts during Sudden Stops: Emerging markets

The following figure complements the analysis in section 2.

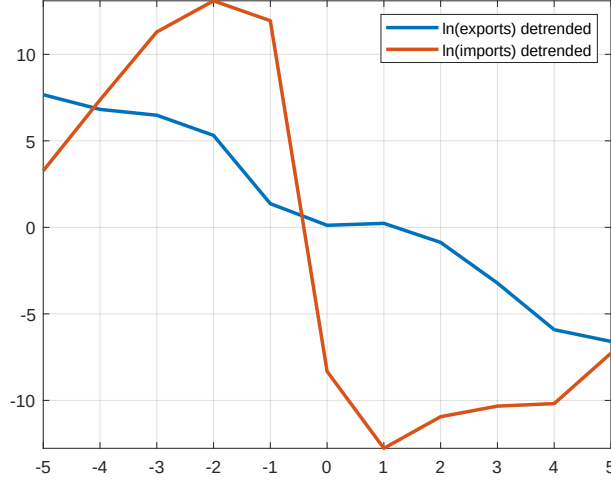


Figure 14: Detrended exports and imports during a Sudden Stop in emerging countries

Note: The period 0 identifies the year of a Sudden Stop episode. The figure presents mean values of percentage deviations of imports and exports from a linear trend in a sample of 59 Sudden Stops.

A.3 Decomposition of the net-exports reversal

Since net exports over GDP is the difference between the export- and import-to-GDP ratios, its change is the exact sum of two contributions,

$$\Delta\left(\frac{X-M}{Y}\right)_t = \underbrace{\left(\frac{X_t}{Y_t} - \frac{X_{t-1}}{Y_{t-1}}\right)}_{\text{export contribution}} - \underbrace{\left(\frac{M_t}{Y_t} - \frac{M_{t-1}}{Y_{t-1}}\right)}_{\text{import contribution}}.$$

Figure 15 reports the average of the two contributions around the Sudden Stop. At impact the reversal, about 7.1 percentage points of GDP, comes mostly from the fall in the imports-to-GDP ratio, about 4.3 percentage points (three-fifths), while the rise in the exports-to-GDP ratio accounts for the remaining 2.9 percentage points (two-fifths). This ratio decomposition, if anything, understates the role of imports: because output falls during a Sudden Stop, the export-to-GDP ratio rises mechanically even when export volumes do not, so part of the export contribution reflects the contraction of GDP rather than an expansion of exports. Consistent with this, in volume terms

real imports collapse by about 25 percent at the crisis while real exports are essentially flat, so the reversal reflects the compression of imports.

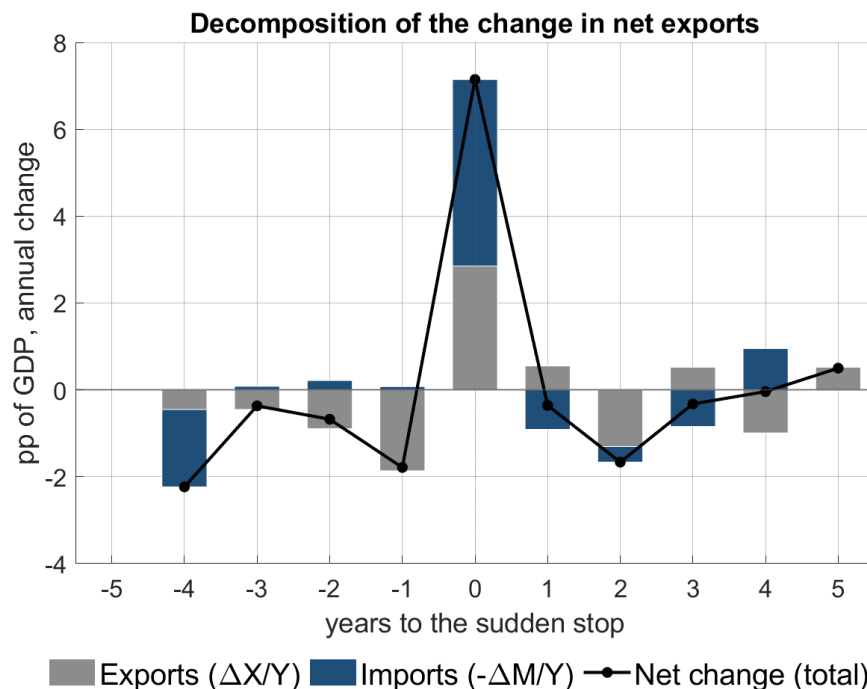


Figure 15: Decomposition of the change in net exports over GDP

Note: Year-over-year change in net exports over GDP around emerging-market Sudden Stops (59 episodes), split into the export contribution $X_t/Y_t - X_{t-1}/Y_{t-1}$ and the import contribution $-(M_t/Y_t - M_{t-1}/Y_{t-1})$. The black line is the total change. Bars in percentage points of GDP.

A.4 Decomposition of the change in total imports

We complement the net-exports decomposition above by splitting the year-over-year change in total imports around a Sudden Stop into the contribution of imported capital goods and that of all other imports. Figure 16 reports it. Although imported capital represents between about 13 and 29 percent of imports across our sample, it falls more than proportionally at the crisis and accounts for roughly 30 percent of the total-import collapse in the Sudden Stop year.

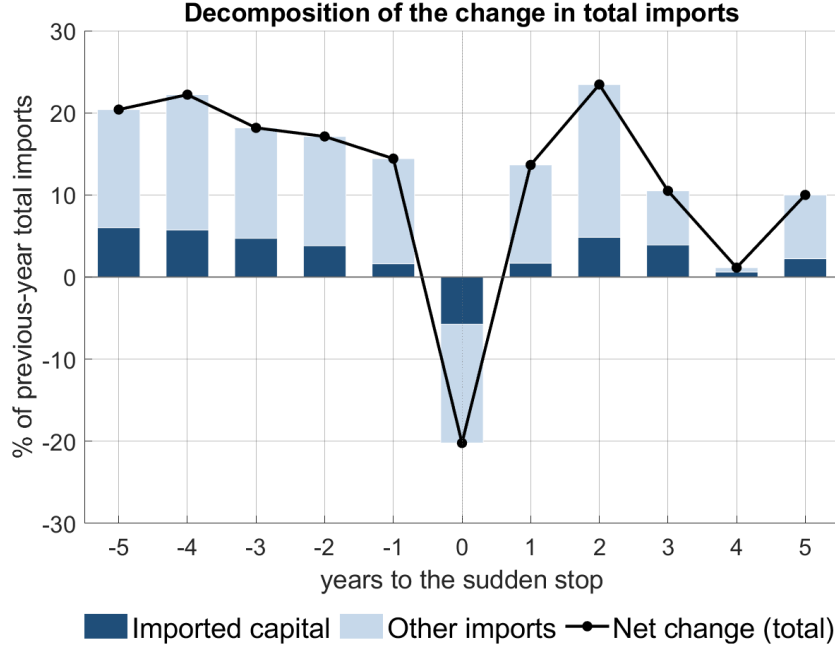


Figure 16: Contribution of imported capital versus other imports to the change in total imports

Note: Year-over-year contribution to the change in total imports around emerging-market Sudden Stops (43 episodes with imported-capital data), as a percent of the previous year's total imports. Each bar splits the change into an imported-capital component and all other imports, from the same spliced UN Comtrade and CEPII BACI source used throughout the paper, so the two sum to the total change (black line).

B Model derivations

This appendix collects the derivations referenced in Section 3.

B.1 The entrepreneur's optimal contract

This subsection derives the optimality condition (11) of the main text. The contract between an entrepreneur and the risk-neutral foreign lender specifies a loan rate, a default threshold, and a leverage ratio, expressed through the functions Γ and G introduced in the Entrepreneurs subsection. Using Θ for the standard Normal CDF,

G admits the closed form

$$G(\bar{\omega}_{j,t+1}^N, \sigma_{\omega,t}^j) = 1 - \Theta \left(\frac{\frac{1}{2}\sigma_{\omega,t}^{2,j} - \log \bar{\omega}_{j,t+1}^N}{\sigma_{\omega,t}^j} \right).$$

Let $R_j^{N,l}$ be the loan return that gives the risk-neutral lender zero expected profits. Entrepreneurs with productivity below the threshold default and the rest repay. In default the lender pays the monitoring cost and seizes the remaining assets. The zero-profit condition is

$$[1 - F(\bar{\omega}_{j,t+1}^N)]R_{j,t}^{N,l}B_{j,t+1}^N + (1 - \mu_{kj}) \int_0^{\bar{\omega}_{j,t+1}^N} \omega_j^N dF(\omega_j^N) R_{j,t+1} q_{j,t} K_{j,t+1}^N = R_{t+1} B_{j,t+1}^N, \quad (22)$$

where $1 - \mu_{kj}$ is the fraction of the return the lender recovers in default and the right-hand side is the cost of the funds raised. The default threshold $\bar{\omega}_{j,t+1}^N$ is defined by

$$R_{j,t+1}^{N,l} B_{j,t+1}^N = \bar{\omega}_{j,t+1}^N R_{j,t+1} q_{j,t} K_{j,t+1}^N,$$

so that below $\bar{\omega}_{j,t+1}^N$ the return on the purchased capital is insufficient to repay the loan. Substituting the definitions of Γ and G , the zero-profit condition can be written in terms of the threshold rather than the loan rate,

$$\frac{R_{j,t+1}}{R_{t+1}} \left[\Gamma(\bar{\omega}_{j,t+1}^N, \sigma_{\omega,t}^j) - \mu_{kj} G(\bar{\omega}_{j,t+1}^N, \sigma_{\omega,t}^j) \right] q_{j,t} K_{j,t+1}^N = B_{j,t+1}^N. \quad (23)$$

The entrepreneur chooses the leverage ratio $\varsigma_{j,t}^N = B_{j,t+1}^N / N_{j,t+1}$ and the default threshold to maximize expected net worth subject to (23):

$$\begin{aligned} & \max_{\varsigma_{j,t}^N, \bar{\omega}_{j,t+1}^N} \mathbb{E}_t \left\{ \frac{R_{j,t+1}}{R_{t+1}} (1 - \Gamma(\bar{\omega}_{j,t+1}^N, \sigma_{\omega,t}^j)) (1 + \varsigma_{j,t}^N) \right. \\ & \left. + \eta_{j,t} \left[\frac{R_{j,t+1}}{R_{t+1}} \left[\Gamma(\bar{\omega}_{j,t+1}^N, \sigma_{\omega,t}^j) - \mu_{kj} G(\bar{\omega}_{j,t+1}^N, \sigma_{\omega,t}^j) \right] (1 + \varsigma_{j,t}^N) - \varsigma_{j,t}^N \right] \right\}. \end{aligned}$$

The first-order conditions deliver the optimality condition (11) reported in the main text.

B.2 Consolidation of the resource constraint

We obtain the aggregate resource constraint (20) by substituting the producers' profits and the entrepreneurial transfers into the household budget constraint (2) and imposing the equilibrium factor-payment conditions.

The domestic investment producer and the two capital producers earn

$$\Pi_{Id,t} = p_{d,t}I_{d,t} - W_{d,t}h_{d,t}, \quad \Pi_{kj,t} = q_{j,t}K_{j,t+1} - q_{j,t}K_{j,t}(1 - \delta_{kj}) - P_{j,t}I_{j,t}, \quad j = \{i, d\}.$$

The net transfer from entrepreneurs to households in sector j is $T_{j,t} = V_{j,t} - N_{j,t+1}$, with $V_{j,t}$ given by (14). Using the return identity $R_{j,t}q_{j,t-1}K_{j,t} = (r_{j,t} + q_{j,t}(1 - \delta_{kj}))K_{j,t}$ implied by (10), the financing identity $N_{j,t+1} = q_{j,t}K_{j,t+1} - B_{j,t+1}$ from (9), and denoting the resource cost of monitoring by

$$M_{j,t} \equiv \mu_{kj} \int_0^{\bar{\omega}_{j,t}} \omega dF(\omega) R_{j,t}q_{j,t-1}K_{j,t},$$

the capital producer's profit and the transfer collapse, once the $q_{j,t}(1 - \delta_{kj})K_{j,t}$ and $q_{j,t}K_{j,t+1}$ terms cancel, to

$$\Pi_{kj,t} + T_{j,t} = r_{j,t}K_{j,t} - P_{j,t}I_{j,t} - R_t B_{j,t} + B_{j,t+1} - M_{j,t}.$$

Substituting $\Lambda_t = \Pi_{ki,t} + \Pi_{kd,t} + \Pi_{Id,t} + T_{i,t} + T_{d,t} - S_t$ into (2) and summing across agents, the labor payment $W_{d,t}h_{d,t}$ cancels with $\Pi_{Id,t}$, leaving the domestic-investment value $p_{d,t}I_{d,t}$, which in turn offsets the term $-P_{d,t}I_{d,t}$ in the sectoral sum since $P_{d,t} = p_{d,t}$. The remaining factor payments vanish through the final goods firm's zero-profit condition,

$$Y_t = r_{i,t}K_{i,t} + r_{d,t}K_{d,t} + W_{f,t}h_{f,t}.$$

Collecting the cross-border flows into the trade balance,

$$TB_t = R_t D_t - D_{t+1} + R_t(B_{i,t} + B_{d,t}) - (B_{i,t+1} + B_{d,t+1}),$$

the household budget constraint reduces to

$$C_t + P_{i,t}I_{i,t} + S_t + TB_t = Y_t - (M_{i,t} + M_{d,t}).$$

Adding $p_{d,t}I_{d,t}$ to both sides and using the definition of measured output, $GDP_t = Y_t + p_{d,t}I_{d,t} - M_{i,t} - M_{d,t}$, yields the resource constraint (20).

B.3 Equilibrium conditions

B.3.1 Equilibrium equations

Household's problem:

$$\begin{aligned} \nu_t \left(C_t - \alpha \tilde{C}_{t-1} - X_{t-1} \frac{h_{f,t}^{\omega_f}}{\omega_f} - X_{t-1} \frac{h_{d,t}^{\omega_d}}{\omega_d} \right)^{-\sigma} &= \lambda_t X_{t-1}^{-\sigma}, \\ \nu_t \left(C_t - \alpha \tilde{C}_{t-1} - X_{t-1} \frac{h_{f,t}^{\omega_f}}{\omega_f} - X_{t-1} \frac{h_{d,t}^{\omega_d}}{\omega_d} \right)^{-\sigma} X_{t-1} h_{f,t}^{\omega_f-1} &= W_{f,t} \lambda_t X_{t-1}^{-\sigma}, \\ \nu_t \left(C_t - \alpha \tilde{C}_{t-1} - X_{t-1} \frac{h_{f,t}^{\omega_f}}{\omega_f} - X_{t-1} \frac{h_{d,t}^{\omega_d}}{\omega_d} \right)^{-\sigma} X_{t-1} h_{d,t}^{\omega_d-1} &= W_{d,t} \lambda_t X_{t-1}^{-\sigma}, \end{aligned}$$

$$\lambda_t = \beta g_{x,t}^{-\sigma} R_{t+1} \mathbb{E}_t [\lambda_{t+1}],$$

$$C_t + D_t R_t = W_{d,t} h_{d,t} + W_{f,t} h_{f,t} + D_{t+1} + \Lambda_t,$$

$$\Lambda_t = \Pi_{ki,t} + \Pi_{kd,t} + \Pi_{Id,t} + T_{i,t} + T_{d,t} - S_t.$$

Final goods producer:

$$r_{d,t} = a_t (1 - \gamma) (X_t h_{f,t})^\gamma K_t^{1-\gamma-\mu_1} a_1 K_{d,t}^{\mu_1-1},$$

$$W_{f,t} = a_t \gamma K_t^{1-\gamma} (X_t h_{f,t})^{\gamma-1} X_t,$$

$$r_{i,t} = a_t (1 - \gamma) (X_t h_{f,t})^\gamma K_t^{1-\gamma-\mu_1} (1 - a_1) \Xi_{t-1}^{\mu_1} K_{i,t}^{\mu_1-1},$$

$$Y_t = a_t (X_t h_{f,t})^\gamma K_t^{1-\gamma},$$

$$K_t = \left(a_1 K_{d,t}^{\mu_1} + (1 - a_1) (\Xi_{t-1} K_{i,t})^{\mu_1} \right)^{\frac{1}{\mu_1}}.$$

Imported capital producer:

$$\begin{aligned}
q_{i,t} - \Lambda_{i,t} \left[1 + \Phi'_{K_{i,t+1}} \left(\frac{K_{i,t+1}}{K_{i,t}} \right) \right] &= \\
\mathbb{E}_t \beta \frac{\lambda_{t+1}}{\lambda_t} g_{x,t}^{-\sigma} \left(q_{i,t+1} (1 - \delta_{ki}) - \Lambda_{i,t+1} \left[(1 - \delta_{ki}) - \Phi \left(\frac{K_{i,t+2}}{K_{i,t+1}} \right) + \Phi'_{K_{i,t+1}} \left(\frac{K_{i,t+2}}{K_{i,t+1}} \right) \frac{K_{i,t+2}}{K_{i,t+1}} \right] \right), \\
\Lambda_{i,t} &= P_{i,t}, \\
K_{i,t+1} &= K_{i,t} (1 - \delta_{ki}) + I_{i,t} - \Phi \left(\frac{K_{i,t+1}}{K_{i,t}} \right) K_{i,t}, \\
\Pi_{ki,t} &= q_{i,t} K_{i,t+1} - q_{i,t} K_{i,t} (1 - \delta_{ki}) - P_{i,t} I_{i,t}.
\end{aligned}$$

Domestic capital producers:

$$\begin{aligned}
q_{d,t} - \Lambda_{d,t} \left[1 + \Phi'_{K_{d,t+1}} \left(\frac{K_{d,t+1}}{K_{d,t}} \right) \right] &= \\
\mathbb{E}_t \beta \frac{\lambda_{t+1}}{\lambda_t} g_{x,t}^{-\sigma} \left(q_{d,t+1} (1 - \delta_{kd}) - \Lambda_{d,t+1} \left[(1 - \delta_{kd}) - \Phi \left(\frac{K_{d,t+2}}{K_{d,t+1}} \right) + \Phi'_{K_{d,t+1}} \left(\frac{K_{d,t+2}}{K_{d,t+1}} \right) \frac{K_{d,t+2}}{K_{d,t+1}} \right] \right), \\
\Lambda_{d,t} &= p_{d,t}, \\
K_{d,t+1} &= K_{d,t} (1 - \delta_{kd}) + I_{d,t} - \Phi \left(\frac{K_{d,t+1}}{K_{d,t}} \right) K_{d,t}, \\
\Pi_{kd,t} &= q_{d,t} K_{d,t+1} - q_{d,t} K_{d,t} (1 - \delta_{kd}) - p_{d,t} I_{d,t}.
\end{aligned}$$

Imported capital entrepreneurs:

$$\begin{aligned}
R_{i,t+1} &= \frac{r_{i,t+1} + q_{i,t+1} (1 - \delta_{ki})}{q_{i,t}}, \\
\frac{R_{i,t+1}}{R_{t+1}} \left[\Gamma(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i) - \mu_{ki} G(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i) \right] (1 + \varsigma_{i,t}) &= \varsigma_{i,t}, \\
\mathbb{E}_t \left[\frac{R_{i,t+1} q_{i,t} K_{i,t+1}}{R_{t+1} N_{i,t+1}} (1 - \Gamma(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i)) \right] &= \\
\mathbb{E}_t \left[\frac{1 - F(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i)}{1 - F(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i) - \mu_{ki} \bar{\omega}_{i,t+1} F_{\omega}(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i)} \right] &= \\
q_{i,t} K_{i,t+1} &= N_{i,t+1} + B_{i,t+1},
\end{aligned}$$

$$N_{i,t+1} = \gamma^e \left[R_{i,t} q_{i,t-1} K_{i,t} - R_t B_{i,t} - \mu_{ki} \int_0^{\bar{\omega}_{i,t}} \omega dF(\omega) R_{i,t} q_{i,t-1} K_{i,t} \right] + W E_{i,t}.$$

Domestic capital entrepreneurs:

$$R_{d,t+1} = \frac{r_{d,t+1} + q_{d,t+1}(1 - \delta_{kd})}{q_{d,t}}$$

$$\frac{R_{d,t+1}}{R_{t+1}} \left[\Gamma(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d) - \mu_{kd} G(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d) \right] (1 + \varsigma_{d,t}) = \varsigma_{d,t},$$

$$\begin{aligned} & \mathbb{E}_t \left[\frac{R_{d,t+1} q_{d,t} K_{d,t+1}}{R_{t+1} N_{d,t+1}} (1 - \Gamma(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d)) \right] = \\ & \mathbb{E}_t \left[\frac{1 - F(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d)}{1 - F(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d) - \mu_{kd} \bar{\omega}_{d,t+1} F_{\omega}(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d)} \right] \end{aligned}$$

$$q_{d,t} K_{d,t+1} = N_{d,t+1} + B_{d,t+1},$$

$$N_{d,t+1} = \gamma^e \left[R_{d,t} q_{d,t-1} K_{d,t} - R_t B_{d,t} - \mu_{kd} \int_0^{\bar{\omega}_{d,t}} \omega dF(\omega) R_{d,t} q_{d,t-1} K_{d,t} \right] + W E_{d,t}.$$

Domestic Investment producer:

$$I_{d,t} = \tilde{a}_t^d (h_{d,t})^\rho,$$

$$\Pi_{I,d,t} = p_{d,t} \tilde{a}_t^d (h_{d,t})^\rho - W_{d,t} h_{d,t},$$

$$W_{d,t} = \rho p_{d,t} \frac{I_{d,t}}{h_{d,t}}.$$

Exogenous shock processes:

$$\ln(g_{t+1}/\bar{g}) = \rho_g \ln(g_t/\bar{g}) + \epsilon_{t+1}^g; \quad \epsilon_t^g \sim N(0, \sigma_g^2); \quad |\rho_g| < 1,$$

$$\ln a_{t+1} = \rho_a \ln a_t + \rho_{a,ad} \epsilon_{t+1}^{a,ad} + \epsilon_{t+1}^a; \quad \epsilon_t^a \sim N(0, \sigma_a^2); \quad |\rho_a| < 1,$$

$$\ln a_{d,t+1} = \rho_{ad} \ln a_{d,t} + \rho_{ad,a} \epsilon_{t+1}^{a,ad} + \epsilon_{t+1}^{ad}; \quad \epsilon_t^{ad} \sim N(0, \sigma_{ad}^2); \quad |\rho_{ad}| < 1,$$

$$\ln \nu_{t+1} = \rho_\nu \ln \nu_t + \epsilon_{t+1}^\nu; \quad \epsilon_t^\nu \sim N(0, \sigma_\nu^2); \quad |\rho_\nu| < 1,$$

$$\ln \mu_{t+1} = \rho_\mu \ln \mu_t + \epsilon_{t+1}^\mu; \quad \epsilon_t^\mu \sim N(0, \sigma_\mu^2); \quad |\rho_\mu| < 1,$$

$$\ln R_{f,t+1} = \rho_{R_f} \ln R_{f,t} + \epsilon_{t+1}^{Rf}; \quad \epsilon_t^{Rf} \sim N(0, \sigma_{R_f}^2); \quad |\rho_{R_f}| < 1,$$

$$\begin{aligned}
\ln(p_{i,t+1}/\bar{p}) &= \rho_p \ln(p_{i,t}/\bar{p}) + \epsilon_{t+1}^p; & \epsilon_t^p &\sim N(0, \sigma_p^2); & |\rho_p| &< 1, \\
\ln \sigma_{\omega,t}^i &= (1 - \rho_\sigma^i) \ln \mu_\sigma^i + \rho_\sigma^i \ln \sigma_{\omega,t-1}^i + \eta_\sigma^i \varepsilon_{\sigma,t}; & \epsilon_t^\sigma &\sim N(0, \sigma_\sigma^{2,i}); & |\rho_\sigma^i| &< 1, \\
\ln \sigma_{\omega,t}^d &= (1 - \rho_\sigma^d) \ln \mu_\sigma^d + \rho_\sigma^d \ln \sigma_{\omega,t-1}^d + \eta_\sigma^d \varepsilon_{\sigma,t}; & \epsilon_t^\sigma &\sim N(0, \sigma_\sigma^{2,d}); & |\rho_\sigma^d| &< 1, \\
s_{t+1} &= (1 - \rho_s) \bar{s} y_{t+1} + \rho_s s_t + \epsilon_{t+1}^s; & \epsilon_t^s &\sim N(0, \sigma_s^2); & |\rho_s| &< 1.
\end{aligned}$$

Definitions:

$$\begin{aligned}
R_t &= R_{o,t-1} e^{\mu_t - 1}, \\
R_{o,t} &= R^* + R_{f,t} + \psi_D \left[\exp \left(\tilde{d}_{t+1} + \tilde{b}_{t+1} - (\bar{d} + \bar{b}) \right) - 1 \right] + \\
&\quad \psi_Y \left[\exp(y_t - \bar{y}) - 1 \right], \\
B_t &= B_{i,t} + B_{d,t}, \\
F(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) &= \Theta \left(\frac{\log(\bar{\omega}_{j,t+1}) + \frac{1}{2}(\sigma_{\omega,t}^j)^2}{\sigma_{\omega,t}^j} \right) \text{ for } j = i, d, \\
G(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) &= 1 - \Theta \left(\frac{\frac{1}{2}(\sigma_{\omega,t}^j)^2 - \log \bar{\omega}_{j,t+1}}{\sigma_{\omega,t}^j} \right) \text{ for } j = i, d, \\
\Gamma(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) &= \bar{\omega}_{j,t+1} (1 - F(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j)) + G(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) \text{ for } j = i, d, \\
T_{i,t} &= \left(1 - \frac{1}{1 + e^{-\bar{\gamma}^e}} \right) V_{i,t} - w_i^e X_{t-1}, \\
T_{d,t} &= \left(1 - \frac{1}{1 + e^{-\bar{\gamma}^e}} \right) V_{d,t} - w_d^e X_{t-1}, \\
V_{i,t} &= R_{i,t} q_{i,t-1} K_{i,t} - R_t B_{i,t} - \mu_{ki} \int_0^{\bar{\omega}_{i,t}} \omega dF(\omega) R_{i,t} q_{i,t-1} K_{i,t}, \\
V_{d,t} &= R_{d,t} q_{d,t-1} K_{d,t} - R_t B_{d,t} - \mu_{kd} \int_0^{\bar{\omega}_{d,t}} \omega dF(\omega) R_{d,t} q_{d,t-1} K_{d,t}, \\
X_t &= \Gamma_t^\eta \left[\left(a_1 \tilde{K}_{d,t}^{\mu_1} + (1 - a_1) \left(\Xi_{t-1} \tilde{K}_{i,t} \right)^{\mu_1} \right)^{\frac{1}{\mu_1}} \right]^{1-\eta}, \\
TB_t &= R_t D_t - D_{t+1} + R_t (B_{i,t} + B_{d,t}) - (B_{i,t+1} + B_{d,t+1}),
\end{aligned}$$

$$GDP_t = Y_t + p_{d,t}I_{d,t} - \mu_{ki} \int_0^{\bar{\omega}_{i,t}} \omega dF(\omega) R_{i,t} q_{i,t-1} K_{i,t} - \mu_{kd} \int_0^{\bar{\omega}_{d,t}} \omega dF(\omega) R_{d,t} q_{d,t-1} K_{d,t}.$$

B.3.2 Stationary equations

There are three sources of growth in the model: the trend in the imported investment price ($P_{i,t}$) given by Ξ_t , the trend productivity shock Γ_t and the trend of the economy X_t , which depends on capital accumulation and Γ_t . The corresponding growth rates are $g_{\Xi,t}$, g_t characterized by equation 17, and $g_{x,t}$.

In order to solve the model, we need to stationarize the equilibrium equations from section B.3, dividing each variable by the corresponding trend. In general, we adopt the notation of working with capital letters for growing variables and lowercase letters for detrended variables. We make an exception for $q_{i,t}$ and $\Lambda_{i,t}$ whose detrended versions are $\tilde{q}_{i,t}$ and $\tilde{\Lambda}_{i,t}$.

First, from equation 20 we need each component of domestic absorption to grow at the same rate as output, X_{t-1} . In particular, $P_{i,t}I_{i,t}$ has to grow at rate $X_{t-1}g_{\Xi,t}$. This outcome happens only if the trend of imported investment is given by $\frac{X_{t-1}}{\Xi_{t-1}}$ and the growth rate of imported investment is $\frac{g_{x,t}}{g_{\Xi,t}}$.

Notice that this fact is consistent with the data. In the dataset, we see that output, consumption, domestic investment, and the trade balance have a similar growth rate, close to 1 percent on average, in the whole period. This growth rate is $\bar{g} - 1$ in the model. However, the imported investment grows at an average rate of 3.6 percent and the imported investment price grows at -2.4 percent, which means the gross growth rate of imported investment prices is 0.9756 on average what we call $\bar{g}_{\Xi}$. Then we have $\frac{g_{x,t}}{g_{\Xi,t}} = 1.035$, very close to the empirical gross growth rate of imported investment.

Since the price of domestic investment does not grow, the trend in domestic investment, and then, the one in domestic capital, is equal to X_{t-1} .

We follow an analogous approach to find the trend of the rest of the endogenous variables. The complete set of stationary equilibrium equations is the following.

Household's problem:

$$\begin{aligned} \nu_t \left(c_t - \alpha \frac{\tilde{c}_{t-1}}{g_{x,t-1}} - \frac{h_{f,t}^{\omega_f}}{\omega_f} - \frac{h_{d,t}^{\omega_d}}{\omega_d} \right)^{-\sigma} &= \lambda_t, \\ \nu_t \left(c_t - \alpha \frac{\tilde{c}_{t-1}}{g_{x,t-1}} - \frac{h_{f,t}^{\omega_f}}{\omega_f} - \frac{h_{d,t}^{\omega_d}}{\omega_d} \right)^{-\sigma} h_{f,t}^{\omega_f-1} &= w_{f,t} \lambda_t, \end{aligned}$$

$$\nu_t \left(c_t - \alpha \frac{\tilde{c}_{t-1}}{g_{x,t-1}} - \frac{h_{f,t}^{\omega_f}}{\omega_f} - \frac{h_{d,t}^{\omega_d}}{\omega_d} \right)^{-\sigma} h_{d,t}^{\omega_d-1} = w_{d,t} \lambda_t,$$

$$\lambda_t = \beta R_{t+1} g_{x,t}^{-\sigma} \mathbb{E}_t [\lambda_{t+1}],$$

$$c_t + d_t R_t = w_{f,t} h_{f,t} + w_{d,t} h_{d,t} + d_{t+1} g_{x,t} + \tilde{\Lambda}_t,$$

$$\tilde{\Lambda}_t = \pi_{ki,t} + \pi_{kd,t} + \pi_{Id,t} + t_{i,t} + t_{d,t} - s_t.$$

Final goods producer:

$r_{i,t}$ grows at rate Ξ_{t-1} . $W_{f,t}$, K_t and Y_t grow at rate X_{t-1} . $r_{d,t}$ does not grow.

$$r_{d,t} = a_t (1 - \gamma) (g_{x,t} h_{f,t})^\gamma k_t^{1-\mu_1-\gamma} a_1 k_{d,t}^{\mu_1-1},$$

$$w_{f,t} = a_t \gamma k_t^{1-\gamma} (g_{x,t} h_{f,t})^{\gamma-1} g_{x,t},$$

$$r_{i,t} = a_t (1 - \gamma) (g_{x,t} h_{f,t})^\gamma k_t^{1-\mu_1-\gamma} (1 - a_1) k_{i,t}^{\mu_1-1},$$

$$y_t = a_t (g_{x,t} h_{f,t})^\gamma k_t^{1-\gamma},$$

$$k_t = (a_1 k_{d,t}^{\mu_1} + (1 - a_1) k_{i,t}^{\mu_1})^{\frac{1}{\mu_1}}.$$

Imported capital producer:

Here, we have the following: $I_{i,t}$ and $K_{i,t}$ grow at rate: $\frac{X_{t-1}}{\Xi_{t-1}}$. $p_{i,t}$, $q_{i,t}$ and $\Lambda_{i,t}$ grow at rate Ξ_{t-1} . Π_t grows at rate X_{t-1} ,

$$\begin{aligned} & \tilde{q}_{i,t} - \tilde{\Lambda}_{i,t} \left[1 + \Phi'_{k_{i,t+1}} \left(\frac{k_{i,t+1}}{k_{i,t}} \cdot \frac{g_{x,t}}{g_{\Xi,t}} \right) \right] = \\ & \mathbb{E}_t \beta \frac{\lambda_{t+1}}{\lambda_t} \frac{g_{\Xi,t}}{g_{x,t}^\sigma} \left(\tilde{q}_{i,t+1} (1 - \delta_{ki}) - \tilde{\Lambda}_{i,t+1} \left[(1 - \delta_{ki}) - \Phi \left(\frac{k_{i,t+2}}{k_{i,t+1}} \cdot \frac{g_{x,t+1}}{g_{\Xi,t+1}} \right) \right. \right. \\ & \left. \left. + \Phi'_{k_{i,t+1}} \left(\frac{k_{i,t+2}}{k_{i,t+1}} \cdot \frac{g_{x,t+1}}{g_{\Xi,t+1}} \right) \frac{k_{i,t+2}}{k_{i,t+1}} \cdot \frac{g_{x,t+1}}{g_{\Xi,t+1}} \right] \right) \end{aligned}$$

$$\tilde{\Lambda}_{i,t} = p_{i,t},$$

$$k_{i,t+1} \frac{g_{x,t}}{g_{\Xi,t}} = k_{i,t} (1 - \delta_{ki}) + i_{i,t} - \Phi \left(\frac{k_{i,t+1}}{k_{i,t}} \frac{g_{x,t}}{g_{\Xi,t}} \right) k_{i,t},$$

$$\pi_{ki,t} = \tilde{q}_{i,t} k_{i,t+1} \frac{g_{x,t}}{g_{\Xi,t}} - \tilde{q}_{i,t} k_{i,t} (1 - \delta_{ki}) - p_{i,t} i_{i,t}.$$

Domestic capital producers:

$q_{d,t}$, $p_{d,t}$ and $\Lambda_{d,t}$ do not grow. $K_{d,t}$, $I_{d,t}$ and $\Pi_{d,t}$ grow at rate X_{t-1} ,

$$\begin{aligned} q_{d,t} - \Lambda_{d,t} \left[1 + \Phi'_{k_{d,t+1}} \left(\frac{k_{d,t+1}}{k_{d,t}} g_{x,t} \right) \right] = \\ \mathbb{E}_t \beta \frac{\lambda_{t+1}}{\lambda_t} g_{x,t}^{-\sigma} \left(q_{d,t+1} (1 - \delta_{kd}) - \Lambda_{d,t+1} \left[(1 - \delta_{kd}) - \Phi \left(\frac{k_{d,t+2}}{k_{d,t+1}} g_{x,t+1} \right) \right. \right. \\ \left. \left. + \Phi'_{k_{d,t+1}} \left(\frac{k_{d,t+2}}{k_{d,t+1}} g_{x,t+1} \right) \frac{k_{d,t+2}}{k_{d,t+1}} g_{x,t+1} \right] \right) \\ \Lambda_{d,t} = p_{d,t}, \end{aligned}$$

$$k_{d,t+1} g_{x,t} = k_{d,t} (1 - \delta_{kd}) + i_{d,t} - \Phi \left(\frac{k_{d,t+1}}{k_{d,t}} g_{x,t} \right) k_{d,t}$$

$$\pi_{kd,t} = q_{d,t} k_{d,t+1} g_{x,t} - q_{d,t} k_{d,t} (1 - \delta_{kd}) - i_{d,t} p_{d,t}.$$

Imported capital entrepreneurs:

$R_{i,t+1}$ does not grow. N_t , $B_{i,t}$, and $WE_{i,t}$ grow at rate X_{t-1} ,

$$\begin{aligned} \frac{R_{i,t+1}}{g_{\Xi,t}} &= \frac{r_{i,t+1} + \tilde{q}_{i,t+1} (1 - \delta_{ki})}{\tilde{q}_{i,t}}, \\ \frac{R_{i,t+1}}{R_{t+1}} \left[\Gamma(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i) - \mu_{ki} G(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i) \right] (1 + \varsigma_{i,t}) &= \varsigma_{i,t}, \\ \mathbb{E}_t \left[\frac{R_{i,t+1} \tilde{q}_{i,t} k_{i,t+1}}{R_{t+1} n_{i,t+1} g_{\Xi,t}} (1 - \Gamma(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i)) \right] &= \\ \mathbb{E}_t \left[\frac{1 - F(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i)}{1 - F(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i) - \mu_{ki} \bar{\omega}_{i,t+1} F_{\omega}(\bar{\omega}_{i,t+1}, \sigma_{\omega,t}^i)} \right] &= \\ \tilde{q}_{i,t} k_{i,t+1} &= (n_{i,t+1} + b_{i,t+1}) g_{\Xi,t}, \\ n_{i,t+1} g_{x,t} &= \gamma^e \left[R_{i,t} \frac{\tilde{q}_{i,t-1} k_{i,t}}{g_{\Xi,t-1}} \left(1 - \mu_{ki} \int_0^{\bar{\omega}_{i,t}} \omega dF(\omega) \right) - R_t b_{i,t} \right] + w e_{i,t}. \end{aligned}$$

Domestic capital entrepreneurs:

$$\begin{aligned} R_{d,t+1} &= \frac{r_{d,t+1} + q_{d,t+1} (1 - \delta_{kd})}{q_{d,t}}, \\ \frac{R_{d,t+1}}{R_{t+1}} \left[\Gamma(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d) - \mu_{kd} G(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d) \right] (1 + \varsigma_{d,t}) &= \varsigma_{d,t}, \end{aligned}$$

$$\mathbb{E}_t \left[\frac{R_{d,t+1} q_{d,t} k_{d,t+1}}{R_{t+1} n_{d,t+1}} (1 - \Gamma(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d)) \right] =$$

$$\mathbb{E}_t \left[\frac{1 - F(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d)}{1 - F(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d) - \mu_{kd} \bar{\omega}_{d,t+1} F_{\omega}(\bar{\omega}_{d,t+1}, \sigma_{\omega,t}^d)} \right]$$

$$q_{d,t} k_{d,t+1} = n_{d,t+1} + b_{d,t+1},$$

$$n_{d,t+1} g_{x,t} = \gamma^e \left[R_{d,t} q_{d,t-1} k_{d,t} \left(1 - \mu_{kd} \int_0^{\bar{\omega}_{d,t}} \omega dF(\omega) \right) - R_t b_{d,t} \right] + w e_{d,t}.$$

Domestic investment producers:

$$\dot{i}_{d,t} = \bar{a}_d a_{d,t} g_{x,t} (h_{d,t})^\rho,$$

$$\pi_{Id,t} = p_{d,t} \dot{i}_{d,t} - w_{d,t} h_{d,t},$$

$$w_{d,t} = \rho p_{d,t} \frac{\dot{i}_{d,t}}{h_{d,t}}.$$

Exogenous shock processes: see section B.3.

Definitions:

$T_{i,t}$ and $T_{d,t-1}$ grow at rate X_{t-1} .

$$R_t = R_{o,t-1} e^{\mu_t - 1},$$

$$R_{o,t} = R^* + R_{f,t} + \psi_D \left[\exp \left(\tilde{d}_{t+1} + \tilde{b}_{t+1} - (\bar{d} + \bar{b}) \right) - 1 \right] +$$

$$\psi_Y \left[\exp (y_t - \bar{y}) - 1 \right],$$

$$b_t = b_{i,t} + b_{d,t},$$

$$F(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) = \Theta \left(\frac{\log(\bar{\omega}_{j,t+1}) + \frac{1}{2}(\sigma_{\omega,t}^j)^2}{\sigma_{\omega,t}^j} \right) \text{ for } j = i, d,$$

$$G(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) = 1 - \Theta \left(\frac{\frac{1}{2}(\sigma_{\omega,t}^j)^2 - \log \bar{\omega}_{j,t+1}}{\sigma_{\omega,t}^j} \right) \text{ for } j = i, d,$$

$$\Gamma(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) = \bar{\omega}_{j,t+1} (1 - F(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j)) + G(\bar{\omega}_{j,t+1}, \sigma_{\omega,t}^j) \text{ for } j = i, d,$$

$$\begin{aligned}
t_{i,t} &= \left(1 - \frac{1}{1 + e^{-\bar{\gamma}^e}}\right) v_{i,t} - w e_{i,t}, \\
t_{d,t} &= \left(1 - \frac{1}{1 + e^{-\bar{\gamma}^e}}\right) v_{d,t} - w e_{d,t}, \\
v_{i,t} &= R_{i,t} \frac{\tilde{q}_{i,t-1} k_{i,t}}{g_{\Xi,t-1}} - R_t b_{i,t} - \mu_{ki} \int_0^{\bar{\omega}_{i,t}} \omega dF(\omega) R_{i,t} \frac{\tilde{q}_{i,t-1} k_{i,t}}{g_{\Xi,t-1}}, \\
v_{d,t} &= R_{d,t} q_{d,t-1} k_{d,t} - R_t b_{d,t} - \mu_{kd} \int_0^{\bar{\omega}_{d,t}} \omega dF(\omega) R_{d,t} q_{d,t-1} k_{d,t}, \\
g_{x,t} &= g_t^\eta g_{x,t-1}^{1-\eta} \left[(a_1 k_{d,t}^{\mu_1} + (1 - a_1) (k_{i,t})^{\mu_1})^{\frac{1}{\mu_1}} \right]^{1-\eta},
\end{aligned}$$

$$tb_t = R_t d_t - d_{t+1} g_{x,t} + R_t (b_{i,t} + b_{d,t}) - (b_{i,t+1} + b_{d,t+1}) g_{x,t}$$

$$gdp_t = y_t + i_{d,t} p_{d,t} - \mu_{ki} \int_0^{\bar{\omega}_{i,t}} \omega dF(\omega) R_{i,t} \frac{\tilde{q}_{i,t-1} k_{i,t}}{g_{\Xi,t-1}} - \mu_{kd} \int_0^{\bar{\omega}_{d,t}} \omega dF(\omega) R_{d,t} q_{d,t-1} k_{d,t}.$$

B.4 Estimation

Table 8: Priors and estimation results: Measurement errors

	Dist.	Prior				Posterior			
		LB	UB	Mean	s.d.	Mean	Median	10%	90%
g_y	IG			0.001	0.0025	0.00	0.00	0.00	0.00
g_c	IG			0.001	0.0025	0.02	0.02	0.01	0.02
g_{i_d}	IG			0.001	0.0025	0.04	0.04	0.03	0.05
g_{i_i}	IG			0.001	0.0025	0.00	0.00	0.00	0.00
tb_y	IG			0.001	0.0025	0.00	0.00	0.00	0.00
rb_y	IG			0.001	0.0025	0.00	0.00	0.00	0.00
g_{p_i}	IG			0.001	0.0025	0.00	0.00	0.00	0.00
R_f	IG			0.001	0.0025	0.00	0.00	0.00	0.00

Note: Posterior distributions from a random walk Metropolis-Hastings algorithm of 990,000 draws per chain across two chains, discarding the first half of each chain as burn-in.

B.5 Quantitative results: Variance decomposition

In the following table, we present the decomposition of transitory shocks' contribution to the volatility of each observable variable.

Table 9: Variance decomposition, disaggregated shocks (%)

Shock	g_y	g_c	g_{i_d}	g_{i_i}	tb_y	rb_y
a_t	16.9	10.0	3.3	0.7	2.1	2.4
ad_t	0.1	0.3	2.8	0.0	0.9	0.5
$\epsilon^{a,ad}$	63.0	47.8	66.1	2.6	25.6	16.7
Transitory	80.0	58.1	72.2	3.3	28.6	19.6

B.6 The interaction between the trend and financial factors: Supplementary figures

In this section, we present the impulse response function of the interest rate, investment, and debt to a one standard deviation shock in the spread (μ_t), the risk-free rate ($R_{f,t}$), or risk (σ_t^i, σ_t^d), to complement the analysis in section 6.2.

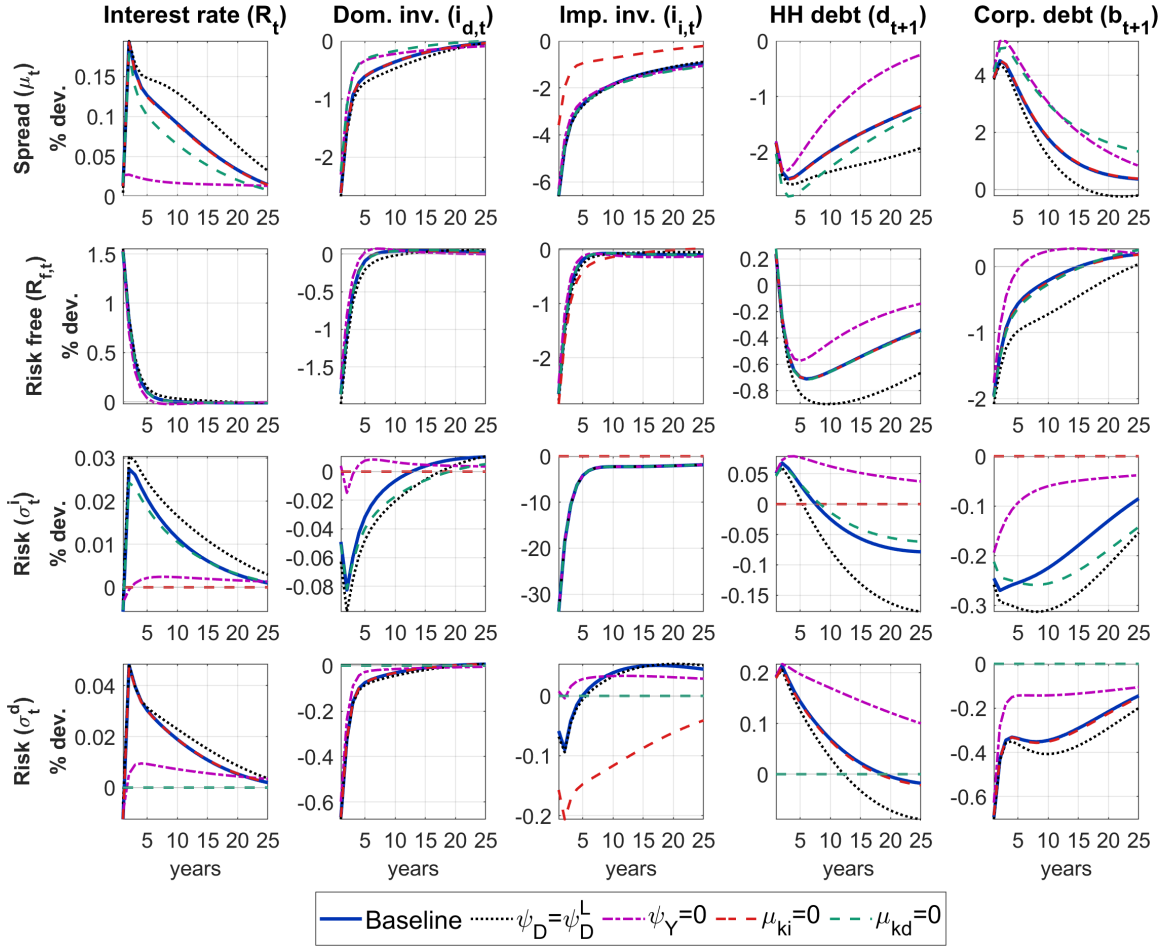


Figure 17: Impulse response function (in %) for the interest rate, investment, and debt

Note: R_t , $i_{d,t}$, $i_{i,t}$, d_{t+1} , and b_{t+1} impulse response function as percentage deviations in % from the steady state to a one standard deviation shock in μ_t , $R_{f,t}$, σ_t^i and σ_t^d .

B.7 Anatomy of Sudden Stops: excluding the 2002 episode

The 2002 crisis is the largest Sudden Stop in the Argentine sample and dominates the cross-episode averages, in particular for imported investment. Figure 18 reproduces the model-versus-data comparison of Figure 9 dropping 2002 from the data. The average impact fall of imported investment in the data shrinks from about 54 to 38 percent, moving toward the model, while the other variables are essentially unchanged. The simulated model moments are the same as in Figure 9.

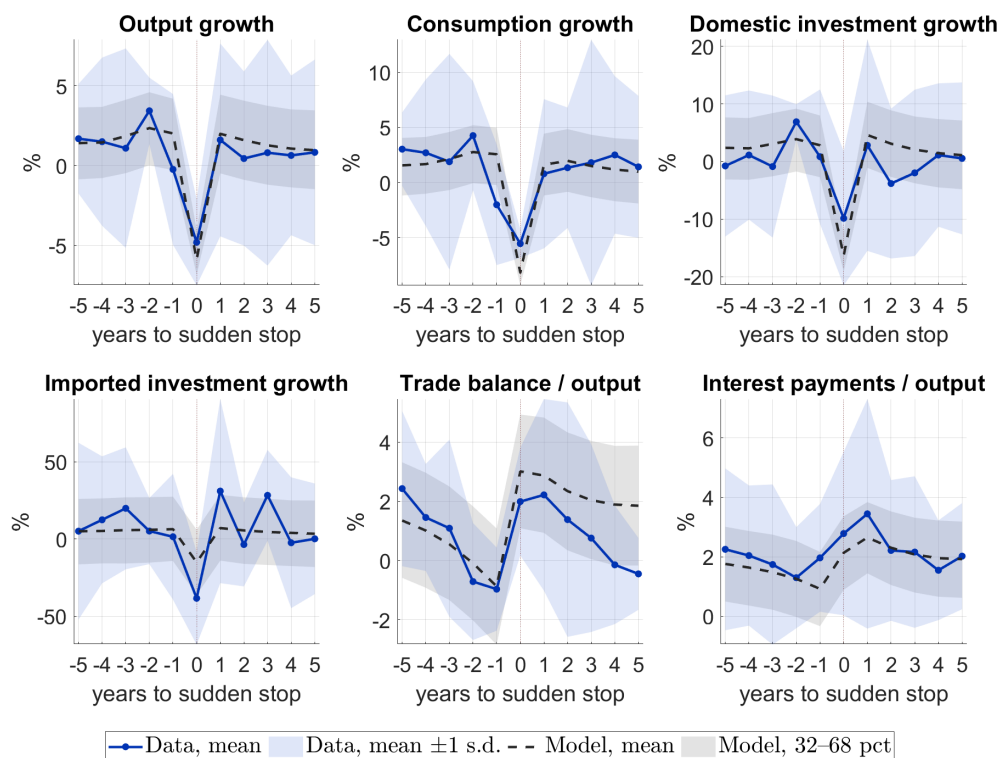


Figure 18: Model versus data around Sudden Stops, excluding the 2002 episode

Note: As Figure 9, but the Argentine data mean and $\pm$ one-standard-deviation band exclude the 2002 episode (five episodes: 1959, 1976, 1982, 1988, 1995). The simulated model moments are unchanged.